PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013

Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

Disaster Preparedness and Response Education System for Schools and Colleges

A PROJECT REPORT

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IN

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

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DECLARATION

We the students of final year **B.Tech** in **INFORMATION SCIENCE AND TECHNOLOGY,** at Presidency University, Bengaluru, named **Varsha B, Mounika M, Mallakaluva Hemanth Reddy,** hereby declare that the project work titled **"Disaster Preparedness and Response Education System for Schools and Colleges"** has been independently carried out by us and submitted in partial fulfilment for the award of the degree of **B.Tech** in **INFORMATION SCIENCE AND TECHNOLOGY,** during the academic year of 2025-26. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

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ABSTRACT

Natural disasters such as earthquakes, floods, cyclones, fires and many others are observed in most parts of the world and may have dire effects of human lives, infrastructure and other economic engagements. This is especially susceptible to countries like India because of the variety of their geography and climatic conditions. Schools and colleges are usually troubled by the problem of equipping students and staff to react favourably whenever there is an emergency. Even though the government authorities offer disaster management guidelines, disaster preparedness education in most institutions is usually cursory in nature such as random safety drills or theoretical awakening programs. The lack of structured training platforms, and simulated learning environments is likely to result in low awareness of students, which may cause confusion and panic in case of a real disaster situation. The Disaster Preparedness and Response Education System is suggested to resolve these issues, as it will position students and the institutional authorities at a higher level of disaster awareness and preparedness. The system will be configured as an online system consisting of a computer-based application, which will incorporate interactive learning modules, simulation of a disaster drill, and a smart chat bot to guide the user on disaster-related information. This platform will help students to study the various disasters, safety measures and evacuation process in an interactive way. The virtual drill element lets users engage in simulated disaster situations, and it makes them aware of how to react in case of disasters. The system has administrative dashboards that enable the institutions to observe training attendance, preparedness levels, and the general awareness on safety in the campus setting. The suggested system is based on the latest web technologies and methods of data management to guarantee the effective work and availability. The chatbot module is an immediate response to disaster-related questions and instructions in the case of emergency. The planned system also facilitates the preparedness evaluation process by the use of performance indicators and participation analytics, and the use of the proposed system by the institutions will provide the evidence of the success in enhancing the efficiency of disaster preparedness training programs among the students in terms of safety awareness. The platform will offer systematic disaster training and create greater awareness of emergency procedures in the learning institutions. Future enhancement can be a combination of real-time alerts of disasters, mobile applications support and location-based disaster response to contribute to disaster management education and campus security.

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ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
AR	Augmented Reality
AWS	Amazon Web Services
CSS	Cascading Style Sheets
DBMS	Database Management System
DPRES	Disaster Preparedness and Response Education System
GCP	Google Cloud Platform
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
ICT	Information and Communication Technology
IoT	Internet of Things
JSON	JavaScript Object Notation
JS	JavaScript
KPI	Key Performance Indicator
LMS	Learning Management System
ML	Machine Learning
NDMA	National Disaster Management Authority
OTP	One-Time Password
PREPWISE	Preparedness Learning and Integrated Safety Education System
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDG	Sustainable Development Goal
SMS	Short Message Service
SQL	Structured Query Language

UI	User Interface
UNDRR	United Nations Office for Disaster Risk Reduction
UX	User Experience
VR	Virtual Reality

CHAPTER 1

INTRODUCTION

The PREPWISE Disaster Preparedness Platform is an online educational platform that aims at enhancing disaster preparedness and awareness among the learners and educational institutions. The site offers interactive tutorials, simulation of emergencies, quizzes and real-time notifications to enable the users learn how to effectively react to a natural disaster like flood, earthquake and cyclone. The system will be designed to ensure that disaster education can be more engaging, more accessible, and more practical by integrating modern web technologies with the assistance features that are smart. PREPWISE helps students, teachers, and administrators to have structured materials to learn, track progress, and disseminate timely safety information to increase disaster preparedness.

1.1 Background

India is the most disaster prone country because of geographical diversity, climate differences and great population density. The country is often hit by natural disasters like earthquakes, floods, cyclones, landslides and fires which have dire consequences on life, infrastructure and economic operations in different parts of the country. Schools and colleges are especially susceptible in these cases since they host huge populations of both students and staff members in the limited space of the campus.

Despite the fact that organizations like the National Disaster Management Authority (NDMA) have developed disaster management structures and guidelines, there is still little disaster preparedness education in the learning institutions. Safety awareness in most instances is accorded either in terms of basic theoretical education or periodical awareness trainings. Practical training or learning through simulation is hardly attentive towards equipping students with skills on how to react well in the event of real emergencies.

The standard of disaster preparedness activities in learning institutions is usually manual, intermittent, and unmonitored. Digital technologies are implemented insufficiently to monitor the level of preparedness, implement organized training programs, or assess the efficacy of the safety awareness programs. This can lead to students and staff being unaware of and suffering the lack of knowledge and confidence to act in the appropriate ways during disaster situations.

As web technologies, cloud computing, and intelligent systems develop, disaster education can be enhanced with the help of the Internet. **The Disaster Preparedness and Response**

Education System in Schools and Colleges is offered with the purpose of offering a more centralized platform of providing the disaster awareness modules, virtual training simulation, emergency communication tool and chatbot-based support in order to promote disaster preparedness and enhance institution safety [1].

1.2 Statistics

The geographical and environmental nature poses a high risk of disaster in India. National Disaster Management Authority (NDMA) records that approximately 58.6 percent of the Indian landmass is prone to earthquakes and approximately 12 percent of the land mass to floods and river erosion. Moreover, about 5,700 km of the Indian coastline is vulnerable to cyclones and storm surges, which often affect the population and infrastructures along the coastlines [2].

According to reports made by the United Nations Office of Disaster Risk Reduction (UNDRR), millions of people each year are impacted by disasters around the world and lose a large amount of money. In third world countries such as India, where schools are a common target of destruction during disasters, the result is a halt in school activities and exposure to student-safety threats. Not with standing these dangers, disaster consciousness among schoolchildren is still a comparatively low level in most academic institutions. Research done in different areas has revealed that a large number of students do not know the emergency evacuation procedures, safe gathering areas, and proper precautions to observe in case of disaster. These statistics demonstrate the necessity to implement disaster preparedness education in institutional learning conditions with the help of the digital system of disaster preparedness education that can offer organized learning opportunities, allow learning disaster preparedness drills, and constantly monitor the level of preparedness among both students and staff.

1.3 Prior Existing Technologies

A number of technological systems have also been established to assist in the management of disasters and their emergency responses. Early warning systems are very common used in detection of natural hazards like earthquakes, floods, and cyclones. These systems make use of sensors, satellite data and meteorological data to detect any potential threats and send messages to law enforcement and communities.

There is also the use of mobile applications and web-based platforms to disseminate information on disaster. These mediums give the citizens safe measures, contact number of emergency departments, and evacuation. GIS is usually employed to map high risk regions and help a government agency plan on how to respond to a disaster.

The majority of technology available, though, is more oriented towards disaster monitoring and disaster early warning than disaster preparedness education. Schools and colleges do not have specific facilities where children can receive organized training courses, interactive learning classes and assessment software.

Thus, a specific digital tool is necessary, aimed at disaster preparedness education in schools and colleges. This type of system can be used to supplement the currently used disaster management technologies with the incorporation of structured training and enhanced awareness amongst the students and the institutional staff.

1.4 Proposed Approach

Aim of the Project

This project seeks to create a digital system of disaster preparedness and response education delivering interactive learning modules, virtual drainage exercises, and smart assistance technologies to promote the level of safety awareness and preparedness at schools and colleges.

Motivation

The growing number of natural catastrophes and absence of organized disaster training in most institutions led to the creation of the system. With the help of modern digital technologies, it is feasible to design an interactive platform that will allow the students to learn disaster response procedures efficiently and acquire feasible preparedness skills.

Proposed Approach

The suggested system is created in the form of a web-based service combining disaster education materials, training based on simulation and real-time communication systems. The system offers interactive modules giving explanations of safety measures in case of various kinds of disasters. Virtual exercises can be performed in which students simulate emergency scenarios and train evacuation protocols and precaution measures.

The site also has the chatbot feature, which offers real-time answers to the disaster-related questions and also offers guidance in emergency cases. Further, an administrative dashboard

enables the institutional authorities to keep track of how people are engaged in training activities and how prepared they are at the institution.

Applications of the Project

The following system may be applied in:

- Educational institutions and schools.
- Colleges and universities
- Disaster awareness training activities.
- Safety education in the community.
- Limit to the Proposed Approach.

The success of the system will depend on the accessibility of internet connection and digital infrastructures in institutions. Also, physical disaster exercises cannot be fully substituted with virtual ones and instead they should be applied as the supplementary training resources to enhance the preparedness.

1.5 Objectives

The key aims of the intended project are the following:

Behaviour:

To create an interactive online system that allows the students to learn the disaster safety protocols via well-organized learning modules and training exercises.

Analysis:

To examine the level of preparedness based on student participation and performance in the disaster awareness modules and simulation of virtual drills.

System Management:

To construct an administrative dashboard that will enable institutions to organize training programs and track disaster preparedness indicators.

Security:

In order to introduce secure authentication systems and provide safe user information management in the disaster preparedness system.

Deployment:

To implement the system as a scalable web-based system that could be available to schools and colleges to learn about disaster preparedness.

1.6 SDGs

The proposed Disaster Preparedness and Response Education System to Schools and Colleges is in line with a number of United Nations Sustainable Development Goals (SDGs) due to the advancement of disaster awareness, education on safety, and community resilience. The SDGs give the world a platform to tackle challenges related to social, environmental, and economic issues and provide sustainable development. Fig 1.1 shows that seventeen Sustainable Development Goals have been adopted by the United Nations, with the main one being SDG 4: Quality Education through the integration of disaster preparedness into the digital learning settings. The project also promotes SDG 11: Sustainable Cities and Communities through interactive modules and the implementation of virtual disaster exercises that help students to be better equipped to response efforts in case of a disaster and to make the campus a safer place. Furthermore, it is in line with SDG 13: Climate Action as it will raise awareness of the disasters and encourage proactive preparedness and mitigation measures in educational institutions.

Figure 1.1 Sustainable Development Goals

SDG 4 – Quality Education

Goal 4, Sustainable Development Goal seeks to provide inclusive and equitable quality education and to enhance lifelong learning opportunities to all. The given system promotes this objective by incorporating the ideas of disaster preparedness training into online learning platforms. The platform helps students to learn the emergency procedures and safety measures in well-organized way through interactive modules, educational materials and virtual disaster exercises.

This system will augment the conventional means of instruction by offering technology-based learning solutions that will augment interaction and learning. Teaching the students about the risks of the disaster and the measures to take to be prepared to cope with it, the platform helps to enhance the overall quality and relevance of the education and to make sure that learners are more equipped to address the challenges of the disaster.

SDG 11 – Sustainable Cities and Communities

SDG 11 is aimed at ensuring the cities and communities are inclusive, safe, resilient, and sustainable. Schools form a significant component of a community and their readiness are critical in minimizing the impact of disasters. The proposed disaster preparedness platform assists schools and colleges to enhance their preparedness with the help of structured training programs and preparedness assessment systems.

The system assists in making campuses less dangerous because it allows institutions to carry out virtual drills and track the level of preparedness. This helps to create disaster resilient communities where people are more prepared to address disasters and reduce risks that may occur.

SDG 13 – Climate Action

Natural disasters like floods, cyclones, extreme weather have been common because of climate change and have been more severe. SDG 13 highlights the urgency of action to be taken to fight climate change and its effects. The current project supports this objective by increasing awareness on the risks of disasters and training students on the right response measures.

The platform promotes awareness of the students about environmental risks and proactive safety measures through digital learning modules and training in the form of simulation. The system facilitates building resilience and adaptive capacity to climate-related disasters by ensuring that people are regularly taught on disaster preparedness.

SDG 3 – Good Health and Well-being

Goal 3 is to guarantee healthy living and life well-being to everyone. The consequences of disasters are usually injuries, psychological pressure, and deaths. These risks can be mitigated by increasing the awareness of disasters and emergency response knowledge.

The suggested system will facilitate this objective by teaching students and staff on the precautionary steps, evacuation, and emergency responses. The platform allows safeguarding the health and well-being of people in educational institutions by training them to act efficiently and calmly in case of a disaster.

1.7 Overview of Project Report

In **Chapter 1**, the author explains the background of the project, statistical information on the occurrence of disasters, and the necessity to have disaster preparedness systems in learning institutions. It also describes the already existing technologies and the intended strategy, project aims, and compliance with Sustainable Development Goals (SDGs).

Chapter 2 illustrates the literature review with regard to the disaster management systems, digital learning platforms, and past research works in disaster preparedness education. It reviews literature and determines gaps that led to design of the proposed system.

Chapter 3 explains the methodology applied to develop the Disaster Preparedness and Response Education System. It outlines the general system architecture, design methodology and the mode of implementation of the software based solution.

Chapter 4 illustrates the project management issues such as project timeline, risk analysis, and project budgeting. It also addresses the planning strategies that have been applied to make sure that the project is completed successfully.

Chapter 5 outlines system analysis and design of the proposed system. It consists of system requirement, block diagram, system flow chart, unit design, standards to be used and specification of domain model. The model of communication, functional view, operational view, and level of deployment of IoT applied in the system architecture are also explained in this chapter.

In **chapter 6,** the hardware and software elements are described to implement the system. It also addresses the software development tools, system implementation of the modules of the software, and simulation of the disaster preparedness system.

Chapter 7 includes the analysis and findings of the suggested system. It explains the test points, test plan, the test results obtained and the lessons learned after the analysis of the system performance.

Chapter 8 The social, legal, ethical, sustainability and safety issues associated with the project are discussed in . It examines the effects of the system on the society, legal and ethical issues, facilitates sustainability, and security in disaster preparedness education.

Chapter 9 provides the conclusion of the project and a general summary of the implementation and results of the Disaster Preparedness and Response Education System. It emphasizes the way it developed platform fulfils the goals of enhancing the disaster awareness and preparedness in educational facilities. The potential future improvement and enhancement of the system are also addressed in the chapter. Moreover, the last part of the report consists of references that were utilized in the study, the original research paper underlining the project, and the appendix with the supporting materials, like the project similarity report, datasets that were used to conduct the analysis, and other materials related to the project.

CHAPTER 2

LITERATURE REVIEW

The background of disaster preparedness education has been growing in terms of the frequency of natural disasters and its necessity as an emergency response system in place. Scientists have investigated and examined some technological interventions like digital learning environments, simulation, mobile applications and intelligent decision-support tools to enhance disaster awareness and preparedness. The chapter provides a review of some of the journal and conference papers concerning the field of disaster management education, emergency response systems, and digital learning technologies.

2.1 Review of Existing Literature

Sharma et al. (2020) offered a web-based disaster management education platform that was aimed at offering awareness training to students and community members. The system incorporates learning resources including safety protocols, disaster management procedures and training videos. The analysis has shown that online platforms are far effective in terms of creating awareness and retaining knowledge among individuals than the conventional classroom lectures. Nonetheless, the system mainly concentrates on theoretical education whereas it lacks interactive simulation modules of training on disaster responses. The authors propose the introduction of interactive drills and real time communication capabilities as future enhancements to make the user more engaged and prepared.

[2] Kumar and Singh (2019) created a mobile application that was meant to enhance disaster awareness and emergency communication in case of natural disasters. The app offers rescue agencies with emergency warnings, safety guidelines and contacts. The study shows how mobile technology is relevant when emergency information is required in a timely manner. The experimental outcomes revealed that users could receive alerts as soon as possible and get access to emergency resources. The system however is very dependent on network connectivity and lacks organized training modules and educational elements to students. The authors are suggesting that learning modules and preparedness assessment tools should be incorporated in future systems.

Patel et al. (2021) introduced a disaster simulation system that aims to train students and emergency responders on virtual scenarios. The suggested system will enable users to be simulated with experiences of disaster events like earthquakes and fires and train on the

relevant safety measures. It was discovered that simulation based learning method enhanced decision making and response time in case of an emergency. Although the system has its benefits, it involves special hardware and high computing capabilities which restrict its accessibility in most learning centres. According to the study, it is proposed to create lightweight simulation systems which can be operated on regular digital platforms.

Lee and Park (2018) examined how disaster education among school learners can be enhanced through gamified learning techniques. The article presented a learning model, which uses a game to educate about the disaster preparedness using quizzes, challenges, and interactive activities. Findings showed that gamification enhances student engagement as well as active learning. Nevertheless, it is a system that is devoted more to individual learning experiences and does not include the features of institutional monitoring and administrative management. The researchers propose the introduction of monitoring dashboards where educators could monitor the progress and the level of participation.

Gupta et al. (2022) developed a smart disaster information platform that integrates disaster-related real-time data collection and emergency communication systems. The system gathers the environmental information and sends warnings to the users when the situation is dangerous. The paper underlines the significance of incorporating real-time knowledge with disaster preparedness training. Though the system offers a good alert and emergency communication platform, it does not have a systematic structure of training students on how to react to disasters before they strike. The authors proposed the use of real-time alert tools with educational tools to enhance disaster preparedness.

As a solution to the challenges, Wang et al. (2019) suggested a cloud-based platform of disaster management that allows institutions, emergency services, and the population to share information. The system is based on cloud technology to store and share information relating to disasters in an efficient manner. The findings show that there is enhanced coordination of the stakeholders during emergencies. Nonetheless, the system is primarily oriented on communication and management of information as opposed to education or training. The novel research directions in the field involve incorporating learning modules and simulation tools into disaster management systems based on the cloud.

Reddy and Prasad (2020) discussed artificial intelligence application in disaster response systems. Their work is devoted to the application of AI algorithms to the disaster-related data analysis and its recommendations of the emergency response strategies. The suggested model increases the speed and accuracy of the decision making in case of a disaster. The system

demands great datasets and sophisticated calculating capacities although it has great technological benefits. The authors note the necessity of simpler AI-based solutions that could be adopted in learning institutions and aid them in disaster awareness and training.

Chen et al. (2021) created a disaster preparedness training platform on the Internet to be used in educational institutions. The site offers well-organized courses, training videos, and healthcare safety measures to students and employees. The results of the evaluation revealed that the platform served a useful purpose in raising awareness and knowledge on disaster safety procedures. Nevertheless, the system does not have the interactive communication capabilities and lacks real-time support to users in the case of an emergency. The authors suggest incorporating chatbot-based support systems to provide disaster-related queries to the users.

[9] Singh and Verma (2018) proposed a community-based model of disaster preparedness, which aims at training the students and the local communities on disaster preparedness using workshops and awareness creation programs. The paper describes the role of community involvement in disaster risk reduction. The strategy is more effective in terms of creating awareness on the local level, yet it entails a lot of human resources and organizational involvement. Rahman et al. (2022) suggested a unified framework of disaster management, which is an integrated platform with education, communication, and monitoring characteristics. The framework contains the training module, emergency alerts, and disaster response coordination modules. According to experimental findings, integrated platforms enhance the level of preparedness and increase coordination of users when facing an emergency. Nonetheless, the complexity of its implementation and its high cost is still an issue in its adoption. The research suggests that simplified digital systems can be developed to be used in educational institutions.

Table 2.1 Synopsis of Literature reviews

Author & Year	**Method /Approach**	**Key findings**
Johnson et al.(2022)	Online disaster education courses	The level of preparedness increased by approximately 65 percent as opposed to the old lecture techniques

Patel and Kumar (2023)	Simulation based disaster training	The virtual simulations not only increased the effectiveness of disaster drills by 78 percent, but also increased the skills of decision making.
Chen et al.(2023)	Techniques of gamification	Gamification provided a 82 percent student engagement and 67 percent knowledge retention
Sharma and Gupta (2022)	Disaster awareness systems on a regional basis	Disaster information that was localized and region specific lowered the emergency response time by 40 percent
Martinez et al.(2023)	Disaster management on cloud-based systems.	The cloud infrastructure offered scalability, high availability
Proposed system	Stimulations, alerts, and chatbot support cloud platform	Interactive learning, virtual exercises, region-specific alerts institutional analytics, and chatbot support of continuous disaster .

2.2 Research Insights and Problem Analysis

Through the reviewed literature, it is evident that a number of technological tools including mobile applications, cloud platforms, simulations tools, and online learning systems have been created to aid in disaster management and emergency response. Such systems are primarily designed to address particular functionalities, including emergency alerts provision, disaster information sharing, or theoretical learning resources. Although the mentioned approaches lead to the better awareness, the majority of the disaster preparedness platforms are not based on the integrated and comprehensive framework including education, practical training, and real-time support within the same platform, which are crucial in enabling students to learn how they may respond in the real-life emergency situation. Moreover, most of the systems in use lack monitoring systems through which institutions can measure the level of preparedness of the students or monitor their participation in the training programs. Such constraints confirm the necessity of a systematic online platform that unites the modules of disaster education, virtual training activities, and monitoring systems to enhance the awareness of preparedness and safety in schools and colleges.

CHAPTER 3

METHODOLOGY

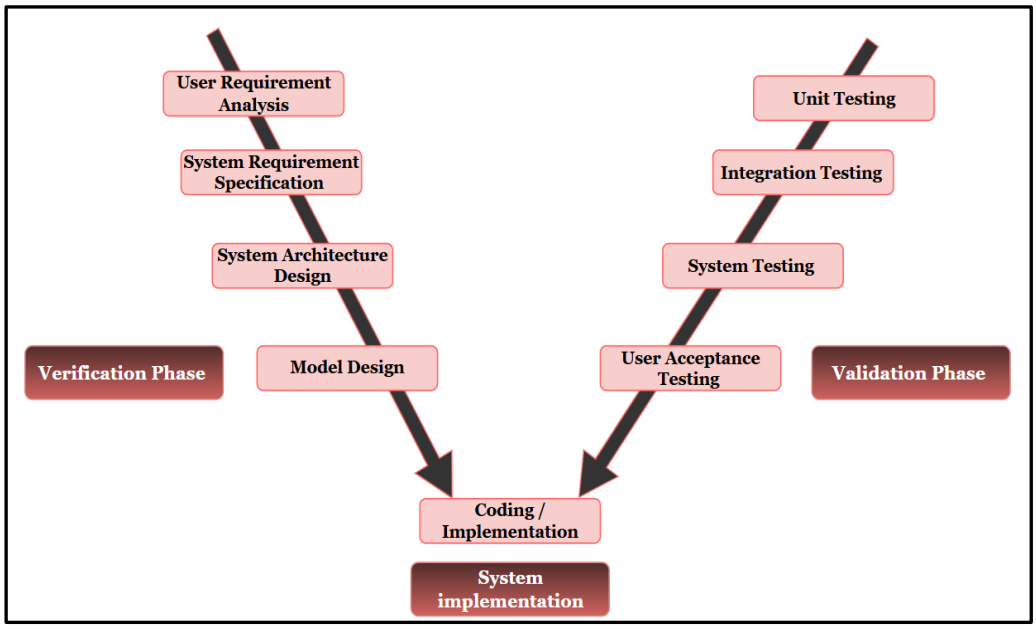


Figure 3.1 V-Model Methodology

The Disaster Preparedness and Response Education System follows V-Model in developing the system. Different development stage in this model is coupled with a testing stage to make sure the system is well verified and validated at each stage of development. The methodology helps in the systematic approach in which, system requirements, design, module development, and implementation are thoroughly thought through before the activity of testing is done. This method assists in detecting the mistakes at the early stages and makes every element of the system work in the manner it was planned, which makes the system more reliable and quality. This methodology will be helpful in the proposed project to design and implement the various disaster education modules, chatbot assistance system, virtual disaster exercises, and administrative monitoring dashboard systematically. This way, the system can be created in a manageable and structured way and at the same time, all the functional components are completely tested, before being deployed.

3.1 Development Approach

To achieve reliability and maintainability, as well as a systematic implementation, the creation of the Disaster Preparedness and Response Education System in Schools and Colleges is developed through a systematic software engineering approach. The project will take the V-

Model development model with its focus on verification and validation throughout the stages of the system development. With the help of this methodology, the project will be developed in distinct stages such as requirement analysis, system design, module development, implementation, and testing.

The systematic format of the V-Model assists in defining design flaws at the initial stage of the development and the fact that every part of the system should be evaluated with the aid of the relevant testing in place. Therefore, the project is confident that the preparedness system regarding the disaster will operate well by using this methodology and achieve the necessary educational and safety goals.

3.2 Requirement Analysis

The initial step in the development process is requirement analysis in which both functional and system requirements of the project are determined. The needs are compiled based on the literature analysis, review of the disaster management guidelines and analysis of the existing disaster education systems.

The key requirements of the proposed system are:

- Creation of an interactive platform of disaster education.
- Virtual disaster drill simulation integration.
- Use of a chatbot to offer disaster instructions.
- Development of a management dashboard to track training.
- User learning data and preparedness scores storage and management.

The implementation of the proposed system and its design are based on these requirements.

3.3 System Design Strategy

Once the system requirements have been identified, the next thing is the system designing. The design stage is associated with the definition of the interaction of various elements of the platform with one another. The suggested system is designed in a modular design approach whereby the system is broken in smaller functional parts.

The key modules of the system are:

- Disaster Education Module
- Disaster Drill Simulation Module Virtual

- Chatbot Assistance Module
- Training Analytics Module
- Administrative Monitoring Module

Working with the backend server and database, each module is planned to execute certain tasks in the system and maintain contact with others.

3.4 System Architecture

The proposed disaster preparedness platform has an architecture that is based on web-based client-server architecture. This architecture will make sure that the users are able to enter the system using web browsers and the data processing and storage activities will be managed by the backend server.

The architecture is made up of the following components:

A. User Interface Layer

Offers interactive learning pages to students.

Training simulation activities and modules.

B. Application Layer

Processes and logic in the system.

C. Database Layer

Stores user information

Sustains progress and preparedness indicators of training.

D. Communication Layer

Allows frontend, backend and database communications.

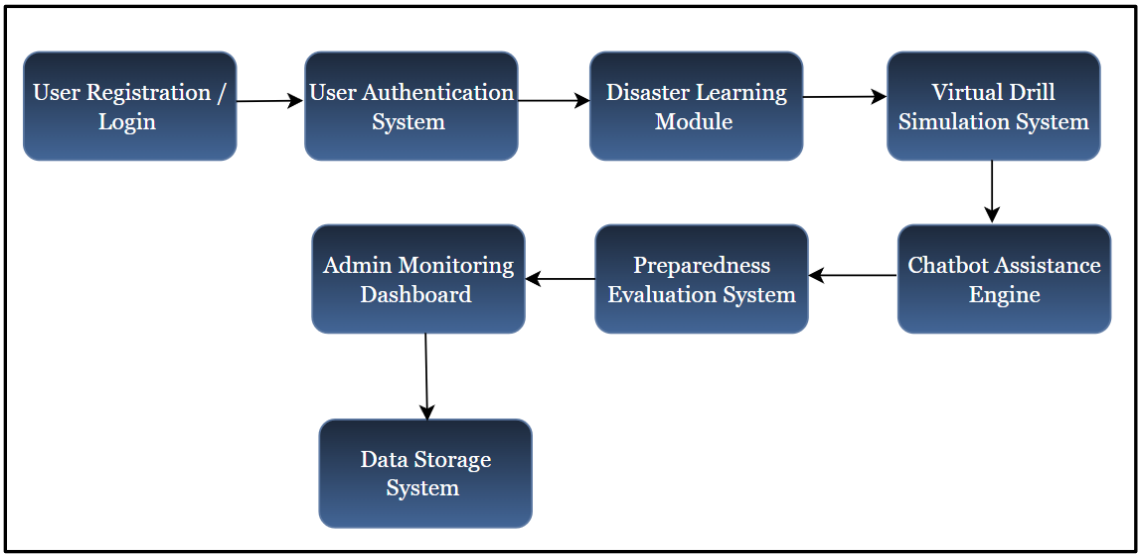


Figure 3.2 System Operation Overview

Fig 3.2 shows the overall working flow of the proposed Disaster Preparedness and Response Education System (DPRES), explaining how different modules are connected and how data moves through the system. The process starts with **User Registration/Login**, where users create an account or log in to access the platform. Once logged in, the **User Authentication System** verifies the user's identity and ensures secure access to the system. After authentication, users enter the **Disaster Learning Module**, where they are provided with educational content about different types of disasters and safety measures. This helps build their basic understanding before moving to practical activities. Next, users proceed to the **Virtual Drill Simulation System**, which allows them to experience simulated emergency situations. This helps them practice how to respond in real-life scenarios. Alongside this, the **Chatbot Assistance Engine** supports users by answering their questions and guiding them during learning and simulations, making the system more interactive and user-friendly. The performance of users in simulations is then analyzed by the **Preparedness Evaluation System**, which measures how well they respond and calculates their preparedness level. These results are sent to the **Admin Monitoring Dashboard**, where administrators can track user progress, evaluate overall preparedness, and monitor system usage. Finally, all user data, performance records, and system information are stored in the **Data Storage System**, ensuring that information is securely maintained and can be accessed when needed. Overall, the diagram represents a complete flow from user entry to learning, practice, evaluation, and data storage, showing how the system works as an integrated platform for disaster preparedness.

3.5 Module Development

The system is broken into several functional modules to make its development simple and increase its maintainability. The modules are put into practice separately and then incorporated to create the whole platform.

A. Disaster Education Module

The module offers a systematic learning material concerning various categories of disaster like earthquakes, floods, fires, and cyclones. The module has safety guidelines, learning materials and interactive activities.

B. Virtual Drill Simulation Module

The simulation module will enable the user to have a virtual practice of the emergency by having simulation drills. These drills enable students to know how they can respond in cases of disasters and how to respond to safety procedures.

C. Chatbot Assistance Module

The chatbot module will help the user to respond to disaster related questions and give directions where there is an emergency situation. The chatbot provides the relevant information based on predefined responses and simple query processing techniques.

Example :

in case user query is equal to earth quake safety:

show safety steps

Elif user query will be a fire emergency:

show evacuation procedure

D. Administrative Monitoring Module.

This module enables the administrators or instructors to monitor the training attendance, measure preparedness indices, and assess the learning student performance.

3.6 System Validation and Testing Approach

Testing is also significant in the process of considering that the system will work as intended and it will address the requirements laid down. The V-Model approach has a testing stage associated with each of the development stages.

The testing stages include:

1.Unit Testing

Each of the modules is tested separately.

The responses of chatbots and functionality of learning modules are checked.

2.Integration Testing

Connection among modules is verified.

Ensures the right communication between chatbot, database and learning modules.

3.System Testing

The complete platform is subjected to system testing.

Checks system functioning, usability and reliability.

4.Acceptance Testing

The platform is tested on by real users, including students and teachers.

Feedback is gathered with the aim of enhancing functionality of systems.

3.7 Implementation Workflow

Project development is a well-organized process of activities, which will guarantee effective execution of the suggested platform.

The following steps are involved in the workflow:

1.System requirements identification.

2.System architecture and modules design.

3.Individual modules development.

4.Merging of modules into one system.

5.System testing and verification.

6.Implementation of the disaster preparedness platform.

This is a systematic process of work, according to which the project is created in a systematic way and addresses the planned goals.

Table 3.1 Stages of Development and activities

Stage of Development	**Major Activities**
Requirement Analysis	the identification of system requirements and user needs.

System Design	Architectural and module design.
Module Development	Implementation of learning modules, chatbot and dashboard.
Integration	Assembling modules into a single system.
Testing	Unit testing, integration and system testing.
Deployment	Making the system accessible to the users.

3.8 Summary

The approach taken to do the project offers a systematic way of coming up with the Disaster Preparedness and Response Education System. The V-Model will be used to ensure that each of the development phases is checked using the development phases, leading to a better system reliability and functioning. The methodology will contribute to the development of a strong digital platform to improve the disaster preparedness education among school and college students by relying on modular design, structured development processes, and systematic validation.

CHAPTER 4
PROJECT MANAGEMENT

4.1 Project Timeline

For the Disaster Preparedness and Response Education System for Schools and Colleges, the project timeline is planned using a Gantt Chart. It helps us clearly understand what tasks need to be done, when they should start, and how long they will take. A Gantt chart is basically a visual chart that shows the complete project schedule in an easy way. It uses horizontal bars to represent different tasks over time. This makes it easier to plan, track progress, and make sure everything is completed on time.

What a Gantt Chart Shows

- **Tasks:**
  It includes all the activities in the project like collecting requirements, designing the system, developing modules, testing, and final deployment.
- **Timeline:**
  It shows the overall project duration, usually divided into weeks.
- **Bars:**
  Each task is shown as a bar. The length of the bar tells how long the task will take, and its position shows when it starts and ends.
- **Dependencies:**
  Some tasks can only start after others are completed. These connections are shown in the chart.
- **Milestones:**
  Important stages like completion of design, development, or final submission are marked.
- **Progress:**
  It helps to track how much work is completed for each task.
- **Responsibility:**
  Tasks can be assigned to team members so that everyone knows their work

How It's Used

- **Planning:**
  The entire project is divided into small and manageable steps.
- **Scheduling:**
  Each task is given a proper timeline so that the work flows smoothly.
- **Tracking Progress:**
  It helps to check if the project is on track or delayed.
- **Managing Work:**
  Work is shared among team members to avoid overload.
- **Communication:**
  Everyone involved in the project can easily understand the plan and progress**.**

The planning phase of the **Disaster Preparedness and Response Education System for Schools and Colleges** focuses on identifying key activities, setting milestones, and defining deadlines. This phase helps in organizing the work properly before starting development.

Table 4.1 summarises the timeline during the project planning phase. The milestones are identified and the deadlines are clearly defined to ensure smooth execution of the project.

Table 4.1 Gantt Chart Data for Project Planning

S. No.	Activity	Description	Duration	Timeline
1	Problem Identification	Define project idea , objectives	1 week	Week 1
2	Requirement Gathering	Collect requirements from users(schools and colleges)	1 week	Week 2

3	Literature Review	Study existing systems And research papers	1 week	Week 3
4	Feasibility Study	Analyse technical, economic feasibility	1 week	Week 4
5	Project Planning	Prepare schedule, resources, and task allocation	1 week	Week 5
6	Proposal Submission	Submit project proposal for approval	1 week	Week 6

The above table clearly shows how the initial phase of the project is structured. Each activity is planned week-by-week to ensure that all requirements are properly understood before moving to the implementation stage

Project implementation

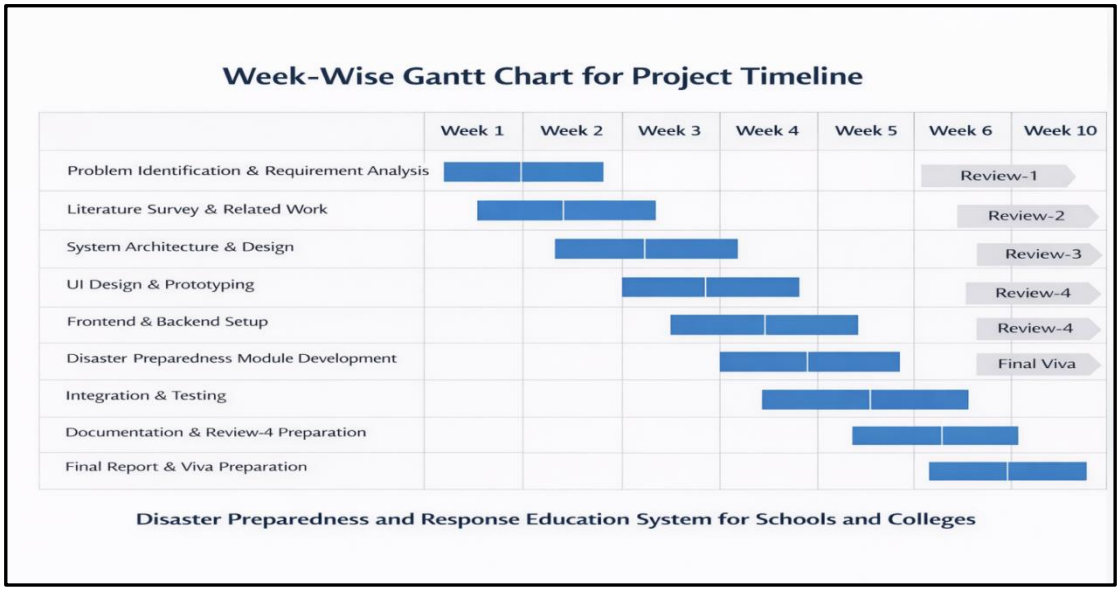


Figure 4.1 Project Implementation Timeline

Fig 4.1 illustrates the detailed project implementation timeline of the proposed Disaster Preparedness and Response Education System (DPRES), organized across multiple weeks. The timeline provides a structured view of how different phases of the project are planned, scheduled, and executed in a sequential and overlapping manner to ensure efficient development. The project begins with the simulation and initial setup phase, which lays the foundation for system development by identifying core functionalities and preparing initial prototypes. This is followed by the User Interface (UI) design phase, where the visual layout and user interaction components of the system are developed to ensure usability and accessibility for students, faculty, and administrators. Subsequently, the integration phase combines various modules such as learning content, simulation engine, chatbot support, and alert system into a unified platform. Alongside this, hardware and infrastructure implementation is carried out to establish cloud storage, database systems, and communication networks required for real-time alerts and analytics. The software development phase spans multiple weeks, focusing on implementing core functionalities including learning modules, disaster simulations, chatbot interactions, and dashboard analytics. Once the development is completed, the project moves into the testing phase, where unit testing, integration testing, and system testing are performed to identify and resolve errors, ensuring system reliability and performance.

Further, the critical evaluation phase assesses the system's effectiveness in terms of usability, preparedness outcomes, and response efficiency. This phase includes

performance analysis and validation of system objectives. In parallel, social, ethical, legal, and sustainability considerations are evaluated to ensure that the platform adheres to data privacy standards, accessibility requirements, and long-term usability.

Finally, the project concludes with report preparation and final documentation, where all findings, results, and system details are compiled into the final research report. The overlapping nature of certain tasks, such as testing and evaluation, indicates a continuous monitoring approach that enhances the overall quality of the system.

Suitability of the Timeline for the Project

The proposed timeline is well-suited for the DPRES system as it ensures a balanced distribution of tasks, efficient time management, and systematic progression from development to deployment. The overlapping of phases such as development, testing, and evaluation allows for early detection of issues and continuous improvement. This structured approach enhances coordination among team members, minimizes delays, and ensures the successful completion of the project within the defined timeframe.

4.2 Project Planning and Team Structure

The development of the **Disaster Preparedness and Response Education System (DPRES)** was carried out using a structured and well-organized project management methodology. The project was divided into multiple phases to ensure systematic execution, proper coordination, and timely completion.

A collaborative team-based approach was adopted, where responsibilities were distributed among team members based on their technical expertise and strengths. This ensured parallel development of system modules and improved overall productivity.

The project team consisted of four members including developers and a project guide who supervised and validated the progress at each stage. Regular communication and coordination among the team members helped in maintaining consistency across modules and achieving milestones effectively.

Team Contribution and Responsibility Distribution

The successful development of the **Disaster Preparedness and Response Education System (DPRES)** was achieved through the collaborative efforts of all team members, where responsibilities were distributed effectively to ensure smooth project execution and timely completion. Each member actively contributed to different aspects of the project including research, design, development, documentation, and presentation.

Mallakaluva Hemanth Reddy (20221IST0007) played a key role in the technical development of the system. He was primarily responsible for coding, implementation of core functionalities, and integration of system modules. He also contributed to the preparation and presentation of project reviews, including Review 1 and Review 2, ensuring that the system development progress was clearly communicated.

Varsha B (20221IST0036) focused on research and documentation aspects of the project. She was actively involved in preparing the research paper, structuring the report, and organizing the overall documentation. Her contribution ensured that the project was well-presented in terms of technical writing and academic standards.

Mounika M (20221IST0034) contributed significantly to the report preparation and documentation process. She supported the development of various sections of the report, ensuring clarity, consistency, and completeness of the content. Her involvement helped in maintaining the overall quality and structure of the documentation.

Throughout the project, all team members collaborated effectively, participated in discussions, and contributed to decision-making processes. This collective effort ensured proper coordination between development and documentation activities, leading to the successful completion of the DPRES system.

4.3 Risk Analysis

Risk analysis is an essential component of project management that helps in identifying potential challenges that may affect the successful development and deployment of the Disaster Preparedness and Response Education System (DPRES). By analyzing possible risks at an early stage, appropriate mitigation strategies can be planned to minimize their impact. One of the major risks associated with the project is technical risk, which includes system failures, software bugs, or integration issues between

different modules such as the simulation engine, chatbot, and database. To address this, proper testing strategies, debugging practices, and modular development approaches are adopted. Another significant risk is connectivity and infrastructure limitations, especially in rural or underdeveloped areas where internet access may be unreliable. Since the system is cloud-based, poor connectivity can affect real-time learning and alert delivery.

This risk can be reduced by optimizing the platform for low bandwidth usage and considering offline support features in future enhancements. User adoption risk is also important, as the effectiveness of the system depends on active participation from students and faculty. Lack of interest or resistance to new technology may limit system usage. To mitigate this, the platform includes interactive modules, gamification elements, and user-friendly interfaces to improve engagement. Data security and privacy risks are critical since the system stores user data, performance records, and institutional information. Unauthorized access or data breaches can affect system reliability. This risk is handled by implementing secure authentication, encryption techniques, and role-based access control mechanisms. Additionally,

dependency on external data sources for disaster alerts poses a risk. Any delay or inaccuracy in third-party data can impact the effectiveness of real-time notifications. Regular validation and use of reliable data sources help reduce this issue.

Finally, project management risks such as delays in development, improper planning, or resource constraints may affect the project timeline. These risks are managed through proper scheduling, continuous monitoring, and effective team coordination. Overall, the identification and management of these risks ensure that the DPRES system is developed in a reliable, secure, and efficient manner, contributing to its successful implementation in educational institutions. Moreover, scalability risk may arise when the system is deployed across multiple institutions with a large number of users accessing the platform simultaneously. This can affect system performance and response time. To mitigate this, cloud-based infrastructure and load management techniques are considered.

Another potential risk is maintenance and update challenges, as the system requires continuous updates to incorporate new disaster data, technologies, and user

requirements. Regular system monitoring and scheduled updates help in maintaining system efficiency and relevance. There is also a training and usability risk, where users may face difficulties in understanding or effectively using the platform features. This can be addressed by providing simple user interfaces, guidance modules, and basic training sessions for students and faculty. By addressing these additional risks through proactive planning and management strategies, the overall reliability and sustainability of the DPRES system can be significantly improved.

4.4 PESTEL Analysis of DPRES Platform

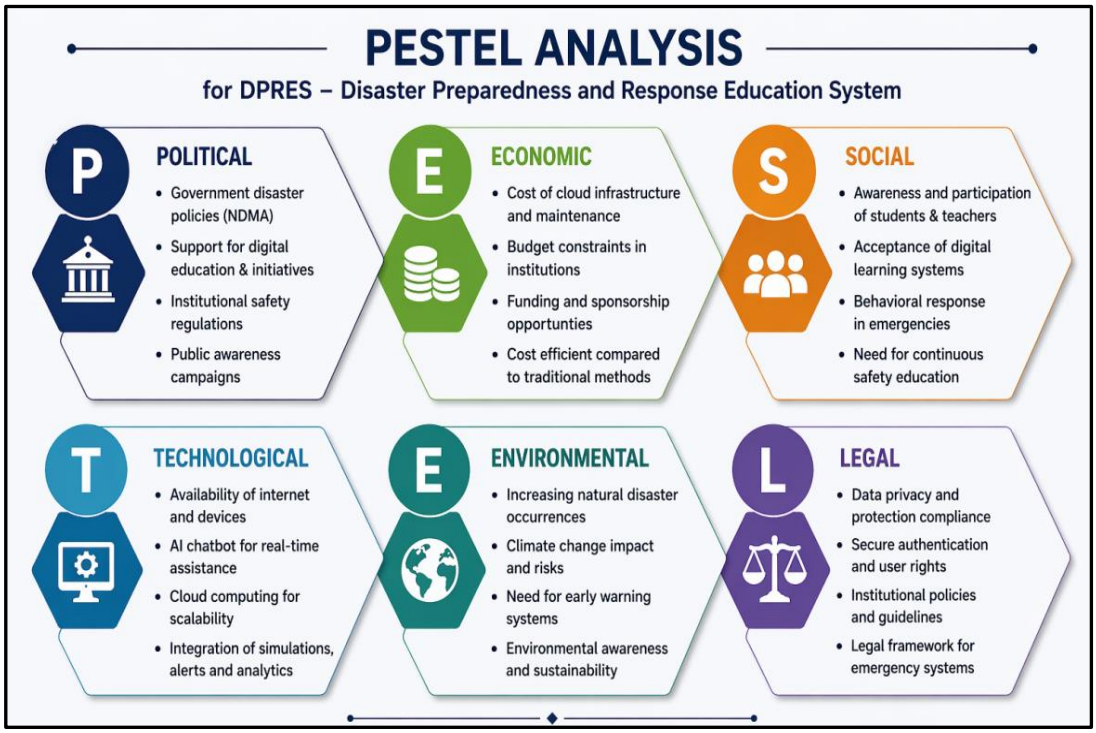


Figure 4.2 PESTEL Analysis of DPRES

The PESTEL analysis is used to evaluate the external factors that may influence the successful implementation of the proposed Disaster Preparedness and Response Education System (DPRES). It helps in identifying potential risks and understanding the broader environment in which the system operates.

- **Political Factors:**
  Government policies, disaster management regulations, and support from authorities such as NDMA play a crucial role in integrating disaster education into academic institutions.

- **Economic Factors:**
  Budget constraints, funding availability, and cost of technological infrastructure may affect adoption, especially in rural or underfunded institutions.
- **Social Factors:**
  Awareness levels, digital literacy, and willingness of students and staff to adopt e-learning platforms influence system effectiveness.
- **Technological Factors:**
  Internet availability, cloud infrastructure, system scalability, and cybersecurity mechanisms are critical for smooth functioning.
- **Environmental Factors:**
  Region-specific disaster risks (floods, earthquakes, cyclones) determine the relevance and customization of training modules.
- **Legal Factors:**
  Data protection laws, user privacy regulations, and compliance with educational standards must be ensured for secure system operation.

4.5 PESTEL Risk Evaluation

Table 4.2 Risk Evaluation based on PESTEL Factors

Factor	Risk Description	Impact Level	Mitigation Strategy
Political	Lack of institutional policy support	Medium	Collaboration with government bodies
Economic	High setup and maintenance cost	High	Use of cloud-based cost-efficient solutions

Social	Low digital awareness among users	Medium	Conduct training and awareness programs
Technological	Poor internet connectivity	High	Offline access and optimized lightweight system
Environmental	Variability of disaster patterns	Medium	Use real-time data APIs
Legal	Data privacy and security issues	High	Implement encryption and secure authentication

4.6 Project Execution Strategy

The DPRES system was developed using a **modular development approach**, where each component of the system was designed, implemented, and tested independently before integration.

The execution strategy included:

- **Phase-wise Development:** Learning modules, simulation engine, chatbot, and alert system were developed separately.
- **Parallel Work Distribution:** Team members worked simultaneously on different modules.
- **Regular Reviews:** Weekly discussions were conducted with the project guide to monitor progress.
- **Integration Testing:** All modules were integrated and tested to ensure smooth communication between components.

This structured approach improved development efficiency and reduced errors during integration.

4.7 Risk Management Table

Table 4.3 Project Risk Management Plan

Risk	Impact	Mitigation
Low user participation	Medium	Awareness programs and gamification
Internet connectivity issues	High	Lightweight system and offline features
Data security risks	High	Encryption and secure login
Integration issues	Medium	Continuous testing and validation
Resource limitations	Medium	Efficient task allocation

4.8 Project Monitoring and Control

Effective monitoring and control mechanisms were implemented throughout the project lifecycle to ensure that the development of the DPRES system progressed according to the planned schedule while maintaining high quality standards. Continuous evaluation

helped in identifying issues early and taking corrective actions to avoid delays and performance gaps.

Monitoring Activities:

1. Weekly Team Meetings

Regular meetings were conducted among team members to discuss project progress, challenges, and upcoming tasks. These meetings ensured effective communication and proper coordination among all members.

2. Progress Tracking Using Timeline

The project timeline (Gantt chart) was used to monitor task completion and ensure that each phase progressed according to the planned schedule.

3. Module-wise Completion Checks

Each module, including the learning system, simulation engine, chatbot, and alert system, was individually monitored to ensure timely completion and correctness.

4. Testing at Each Development Stage

Continuous testing, such as unit testing and integration testing, was performed throughout development to identify and resolve errors at an early stage.

5. Feedback from Project Guide

Periodic reviews with the project guide were conducted to validate progress, improve system design, and ensure that the project met both academic and technical requirements.

Control Measures

1. Fixing Delays through Rescheduling

Any delays identified during monitoring were addressed by revising timelines and redistributing tasks among team members.

2. Debugging and Performance Optimization

Errors identified during testing were promptly fixed, and system performance was optimized to ensure smooth and efficient functioning.

3. Ensuring Consistency Across Modules

Standard coding practices and uniform design principles were maintained across all modules to achieve seamless integration.

4. Quality Assurance Checks

Final verification and validation were performed to ensure that all system functionalities met the required standards and operated reliably.

4.9 Summary

The project management approach adopted for the DPRES system played a crucial role in ensuring the successful completion of the project. A well-structured plan, combined with effective coordination among team members, enabled smooth execution of all project phases from requirement analysis to final deployment. The use of clear role distribution allowed each team member to focus on specific responsibilities, thereby improving efficiency and reducing redundancy in work. The implementation of a detailed project timeline ensured that all tasks were completed within the specified deadlines while maintaining proper workflow management. Continuous monitoring and evaluation throughout the project lifecycle helped in identifying potential issues at an early stage and applying corrective measures promptly. This proactive approach minimized risks and ensured consistent progress. Furthermore, the integration of risk analysis techniques, such as PESTEL analysis and risk management planning, enabled the team to anticipate challenges related to technology, resources, and external factors. Appropriate mitigation strategies ensured that these risks did not significantly impact the project outcome. The collaborative teamwork, supported by regular communication and guidance from the project supervisor, contributed to maintaining quality and consistency across all system modules. The use of modern tools and technologies further enhanced development efficiency and system performance. Overall, the project management strategies adopted for DPRES resulted in the successful development of a scalable, reliable, and efficient disaster preparedness platform. The systematic approach not only ensured timely completion but also strengthened the robustness and usability of the final system, making it suitable for real-world implementation in educational institutions.

CHAPTER 5
ANALYSIS AND DESIGN

The proposed Disaster Preparedness and Response Education System (DPRES) requires analysis and design as critical steps in its process of development. The analysis stage is aimed at comprehending the existing problems in disaster preparedness in the educational facilities and determining the requirements of the system users. The step is associated with research concerning the current disaster management methods and the review of associated literature and the limitations of the existing awareness programs. Most institutions base their training predominantly on theoretical training and safety drills that are conducted few times, which are not sufficient to train students and staffs in the actual emergency scenario. The interactive disaster education and virtual simulation drills and digital training monitoring are the main system requirements determined through the analysis process.

Design stage transforms the requirements identified to a structured and viable system solution. The system architecture is scheduled in this stage, and the platform architecture is determined. The proposed system is billed as a web-based application designed to support interactive learning modules to various forms of disasters, virtual disaster simulation exercises allowing users to train on responding to a disaster, and an intelligent chatbot to guide and respond to disaster related questions. The system also has administrative dashboards which enables institutions to track the participation in the training process and measures the level of preparedness among the students.

Moreover, the design stage is aimed at the fact that the system should be easy to use, scalable, and be readily available on various devices and platforms.

5.1 Requirements

System requirements determine the desired system functionality and performance requirements. These have been divided into functional and non-functional requirements.

System Hardware Requirement Phase

1.The system will need the use of Smartphones, laptops or desktop computers with a reliable internet connection.

2.The users engage with the system through the input of such items as allowing them into the system with the help of their logins, choosing learning modules, and typing in queries to the chatbot.

3.The system provides such outputs as educational material, simulation outcomes, real-time notifications, and chatbot replies.

4.The users devices and system are connected through the network, this provides effective transfer of data.

5.Specifications and speed of devices used on the internet are limiting factors that may determine the system performance.

System Software Requirement Phase

1.The platform is based on web-based system that is underpinned by frontend and backend technologies.

2.The inputs are the user data, disaster-related information, and between modules and chatbot.

3.The frontend will also be connected to the backend server, and the server can also interact with the database and chatbot module to handle requests.

4.The server performance, network latency, and data security and reliability are among the constraints.

Data Collection Requirement :

1.The system gathers the user information, activity logs and interaction records.

2.Data related to disasters, including alerts and information on regions is also gathered.

Data Analysis Requirements :

1.The system evaluates user involvement and learning module and simulation performance.

2.The level of preparedness is measured against the activity and engagement of the users.

System Management Requirement :

1.The system offers an administration board to track the users and trace the general preparedness.

2.It provides ability to control learning content, alerts and updates to the system.

Security Requirements :

1.There are user authentication and authorization policies in place.

2.Data containing sensitive information is safely kept and is not accessed by any third party.

User Interface Requirements :

3.It must have a user-friendly, interactive and simple interface.

4.It ought to facilitate navigation among students, teachers, as well as administrators.

5.The chatbot conversation must be straightforward and easy to follow.

Table 5.1 Overview of requirements

Purpose	To offer a digital disaster preparedness education and response training
Behaviour	Provides interactive modules, simulators, alerts in real-time, and chatbot support.
System Administration	Admin control panel to check user activity and readiness rates.
Data Analysis	monitors the performance of the users and provides insights into the level of preparedness.
Application Deployment	Web based application that can be accessed on numerous devices.
Security	Secure data handling mechanism and user authentication.

5.2 Block diagram

The block diagram of the proposed Disaster Preparedness and Response Education System is a general flow of the system in terms of its functionality.

The input block encompasses user related information, which includes registration information, module choice, simulation input, and chatbot request. It also contains information about disaster like region specific information.

Processing block carries out major activities in the system. It involves user authentication, learning module execution, simulation processing, alerts creation, chatbot communication and data management. This block makes sure that there is a good coordination of various system functionalities.

The output block gives the users the final results. They are interactive learning content, simulation results, real-time notifications, preparedness dashboard, and chatbot replies. The system helps people to obtain relevant and timely information to be prepared in case of a disaster.

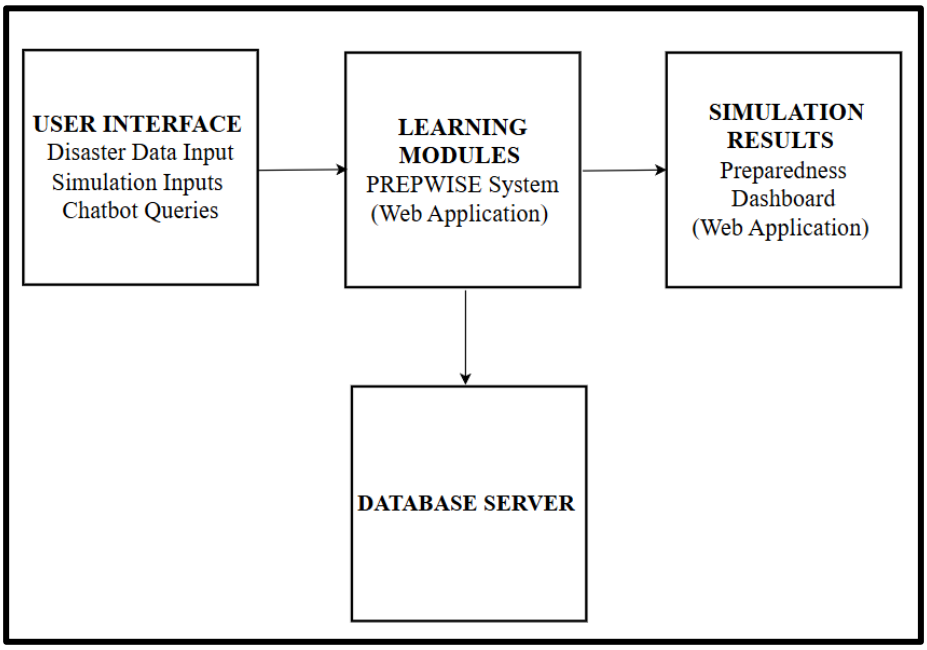


Figure 5.1 System block Diagram

Fig 5.1 Presented is the functional block diagram of the proposed Disaster Preparedness and Response Education System which demonstrates the key components and how they interact in the platform. The major block depicts the PREPWISE web application that receives user requests and organizes the operations of the system. The system receives as inputs the interactions with the user, the information about a disaster, simulation parameters, and the chatbots requests. The processing unit deals with authentication and disaster learning modules, simulations, alert generation and chatbot services. The database server retrieves and stores information by the use of the system. The products of the system are the disaster learning materials, simulation outcomes, preparedness dashboards, alert messages, and chatbot replies that help the users to comprehend the disaster response plans.

5.3 System Flow chart

Disaster Preparedness and Response Education System has a systematic workflow that directs users through various stages of interaction with the site. The system is made to take care of disaster awareness, simulation training and providing emergency information in a systematic way. The workflow will be in place to guarantee that the user can navigate with ease various functionalities which include learning modules, disaster simulation, chatbot support and alert system.

Upon user access, the platform will first authenticate before allowing access to system services. After authentication, users are granted access to role-based access which enables them to interact with modules they need to interact with in the system. Disaster learning resources and simulations are mostly utilized by students, whereas the activities of the system are controlled by the administrators and preparedness programs are monitored.

There are also automated support capabilities like AI-based chatbot and region-based disaster alerts, which are a part of the workflow. These elements assist in making sure that users get access to information and directions in a timely manner in regards to disaster preparedness and safety protocols.

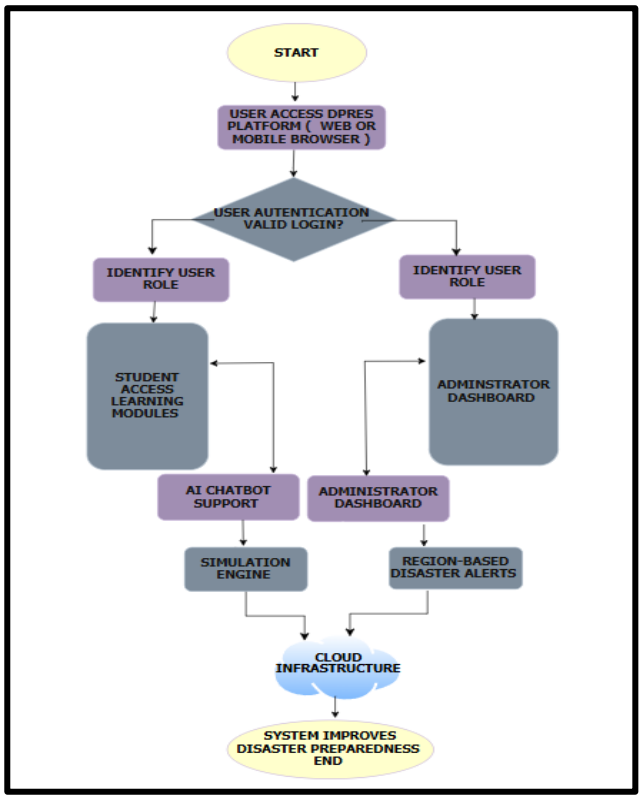


Figure 5.2 System Flowchart of DPRES

The **Fig 5.2** provides the operational workflow of the Disaster Preparedness and Response Education System. It starts when the user can enter the PREPWISE platform via a web browser or a mobile device. This system will then carry out user authentication as a way of checking whether the displayed user login credentials are authentic. Once the authentication has been successful, the role of the user in this platform is then identified. Depending on the assigned role, users are taken to the respective interface. Learning modules are made available to students, and they get access to the contents of disaster preparedness and undergo simulations. They have the ability to communicate with AI chatbot to seek advice regarding disaster safety and response measures. Administrators are redirected to the administrative dashboard where they can observe operations of the platform and control the resources of the system. The simulation engine plays disaster scenes which make users aware of the response strategies they should use in case of an emergency. As well, the system offers disaster notifications based on region to educate users on the possible risks. All the activities of the system are facilitated by the cloud infrastructure that stores and manages user and application data. The process however leads to the ultimate improvement of the disaster awareness and preparedness of the users thus achieving the goal of the system.

Phase Design Hardware Unit Functional phase:

The hardware design phase is concerned with the identification of the elementary computing infrastructure to be used in the running of the Disaster Preparedness and Response Education System. The proposed system is largely a web-based application and so does not use any specialized embedded hardware elements. Rather the system is executed on the basis of user devices which are common and the server infrastructure. The users can access the platform via their desktop, laptop, tablet, smartphones with access to the internet. These gadgets serve as the key interface to the system to access disaster preparedness modules, take part in simulating, and receive alerts. The backend services of the system are run on a server environment that handles the processing of applications, the storage of data and communication of the system. To facilitate compatibility and accessibility, the server architecture will guarantee the successful execution of application services like user authentication, disaster learning modules, chatbot responses, and alert notifications. The platform can be run on standard hardware settings and internet connection without any special equipment and basic

performance testing is performed to make sure that the system works with the various devices used by the end users as well as their browsers. This will provide ease of interaction between user machines and server environment.

Phase Design Software Unit Functional phase:

The software design stage aims at the segmentation of the system into several functional modules that are used to realize the disaster preparedness platform. The system has a number of key software modules that serve a particular purpose and communicate with other modules to facilitate a smooth running of the system. The system has a number of key software modules including user authentication, disaster learning modules, simulation services, chatbot assistance, alert management, and administrative monitoring. These modules are to be used to have a complete environment with disaster preparedness learning. Authentication module is a module that handles the registration and login of the user and only the authorized user should have access to the platform. The learning module entails a background of educational resources associated with calamity recognition and safety measures. The simulation module enables the user to get a virtual feel of what a disaster entails and what strategies should be employed in dealing with it. The chatbot module will give immediate advice by responding to disaster-related queries and helping the user on safety measures. The alert system sends notifications and warnings of a disaster according to the regional disaster data, and an administrative dashboard is also introduced to provide the administrators with the opportunity to manage the resources of the system and monitor the activity of the user and assess the disaster preparedness program. All the software components are tested separately to make sure they work properly and thereafter they are integrated with the general system. Integration testing is then carried out to confirm that there is a smooth communication between modules and to ensure that the entire platform will be reliable to perform.

5.4 Technology Selection and System Environment

The choice of technologies is significant to make the proposed system a web-based application since performance, scalability, and usability are the key factors to consider. The system is intended to be used on standard computing devices without special hardware components.

Selected Platform

PREPWISE uses the web application platform as it enables users to use any device to access the system through a browser. It is flexible, simple to maintain, and it helps with real-time alerts, simulating and chatbot interaction features without the need to install.

References

i. National Disaster Management Authority, India..

ii. UNDRR Guidelines on Disaster risk Reduction.

5.5 Designing Units

PREPWISE system consists of several functional units depending on characteristics one can access upon log-in. The units are constructed to carry out a certain task, so that the system runs smoothly.

1.Dashboard Unit

The dashboard unit gives the user a summary of the system. It shows valuable data including recent activity, alerts and prompts to various modules. It serves as the principal navigation system of the system.

2.Analytics Unit

The analytics department monitors user performance and activity. It evaluates the data including course completion, quiz performance, and simulation attendance to assess the level of preparedness.

3.User Profile Unit

The unit deals with user data, including personal data, progress, and history of activities. It gives the users the opportunity to update their profile and monitor their progress in learning.

4.Learning Unit

The learning module is divided into several sub- modules that offer disaster preparedness education.

Courses: Offers organized learning materials that have to do with various disasters.

Training: Provides practical advice and step by step operation on dealing with emergencies.

Quiz: Can test the knowledge of the user by means of questions and tests.

Tips: Shortcuts to safety tips and best practices.

Certificates: Issues certificates on completion of courses or training programs successfully.

5.Simulation Unit

Through this unit the users can feel like they are in a disaster situation. It guides users through the real life scenario and teaches them how to behave in case of a disaster.

6.Safety Unit

The safety unit offers the much needed information and resources pertaining to safety.

Emergency Kit: includes information on what is needed in a disaster.

Shelters: Displays the safety places or shelters.

Reports: This feature permits viewers to report incidents or access safety related report.

7.Chatbot Unit

The chatbot unit supports the user by responding to their queries concerning disaster preparedness. It gives immediate directions and guides the users in their way through the system.

8.Unit Integration

All the units are tied together and collaborate towards offering a comprehensive disaster preparedness system. The integration also facilitates the flow of data between modules and also operates the platform effectively.

5.6 Standards

The PREPWISE system adheres to the appropriate software and web-based standards in order to provide interoperability, security, reliability, and effective data mixing. The system is a web application and hence the standards are primarily concerned with communication, data handling and security but not the hardware or IoT-specific standards.

Communication and Networking Standards.

The system is run on regular internet communication protocols. The communication between the server and client is done via HTTP and HTTPS. When using HTTPS, data transmission via the network is guaranteed to be secured.

Data Format Standards

The system also employs structured data formats in order to exchange data across modules effectively. JSON data is stored and transferred in a lightweight format; it is easy to read and write due to its lightweight nature.

Security Standards

The system has its security that is vital in securing user data and safe communication. The system adheres to common security measures like encrypted communication with the use of HTTPS and secured authentication mechanisms. They are security best practices as far as web applications are concerned.

Web Standards

The system is developed in a standard web development with the use of such technologies as HTML, CSS, JavaScript. These standards make sure that there is compatibility across various browsers and devices so that the user experience will be consistent.

The Software and System Standards.

The system has been developed according to the general tenets of software engineering principles to make it maintainable, scalable, and reliable. Efficiency in data handling, modularity and structured workflows are exercised to ensure the efficiency of the system.

Benefits of Using Standards

- Makes sure it is compatible with a variety of devices and platforms.
- Enhances security of the system and guards user information.
- Allows the exchange of data among the system components.
- Increases the stability and the performance of the system.
- Scalable and upgraded system in the future.

References

- ISO/IEC 27001- Information Security Management.
- The ISO 9001 - Quality Management Systems.
- Web Standards World Wide Web Consortium (W3C).
- National Disaster Management Authority (NDMA), India

5.7 Map Structure of the PREPWISE System

The disaster preparedness platform PREPWISE can be structured in a layered system structure contributing to the comprehension of how various components of the system interact and fulfil certain tasks. In Layer-based mapping, the system architecture is partitioned into various levels with each layer containing specific form of operation e.g. device interaction, communication, data manipulation, storage and services at the user level. This arrangement provides a simplified manner of analysing the platform displaying the flow of data between user devices through network communication and system processing modules to the end users application services. When the PREPWISE platform is mapped into various functional layers, it would be easier to determine the technologies deployed at each functional level, coordinate the interaction between modules, and provide a smooth interaction of the components of the system. Layered architecture also advances the issue of system scalability and reliability as it has different levels which can manage responsibilities and ensure that disaster preparedness learning modules and simulation systems, alert services and chatbot assistance are effectively handled. The mapping approach gives a clear picture of how PREPWISE system facilitates user interaction, data services and application services to enable an integrated disaster preparedness education platform

Table 5.2 System Layer Mapping

Layer	Reference Architecture Layer	Prepwise System Components	Security Measures.
7	**Processes and Collaboration layer**	Disaster preparedness training,user awareness programs,learning activities	User authentication and role-based access.
6	**Application layer**	PREPWISE dashboard, learning modules, simulator interface,	Secure login and secured access to the application.

		safety services	
5	**Data Abstraction layer**	User progress data process, disaster alert information, analytics generation	Data encrypting and data access control.
4	**Data storage layer**	It stores user account, learning, records, simulation results and alert data	Database security and backup system.
3	**Edge processing layer**	Simulation engine operations, chatbot processing disaster alert handling	Secure data processing and input validation
2	**Connectivity layer**	Connection layer Internet communication, web APIs, HTTP communication between the user and the system	Secure network communication protocols.
1	**Device layer**	Mobile phones, laptops, and desktop computers will be used to access the platform	Device authentication and secure access to the system

5.8 Domain Model Specification

The domain model depicts the primary entities and components, and interactions in the PREPWISE Disaster Preparedness Education System to be used in schools and colleges. It gives us a formal perspective of the interaction among the various elements of the system i.e. students, teachers, administrators, system modules and application services within the platform. The model is aimed at determining the main elements which are needed to prepare to the disaster such as educational modules, disaster

simulation, alerts, and chatbot support. In PREPWISE, students use the system via digital devices to study disaster preparedness, and a teacher and faculty offer academic support and track student progress. The administrator deals with system setup, learning materials and accessibility of the individuals. The domain model assists in clearly defining the relationship among the system components and also makes all the modules to be effective with each other. The domain model enhances clarity in the system, assists in scalability, and aids in the systematic design of the platform by the developers in order to effectively educate about disaster preparedness in the education institutions.

Table 5.3 Description of Domain model

User Entity	Identifies students, teachers, and administrators that will use the PREPWISE disaster preparedness system.
Application Entity	The PREPWISE web application dealing with learning modules, disaster simulations, alerts and chatbot services.
Device Entity	Devices will be mobile phones, laptops or desktops computers that will be used by students and teachers to access the system.
Resource Entity	Contained information: learning materials, disaster information, simulation data and user records obtained in the system database.
Service Entity	System services Authentication service, learning module service, simulation service, alert system, and chatbot support that facilitate the platform to operate accordingly.

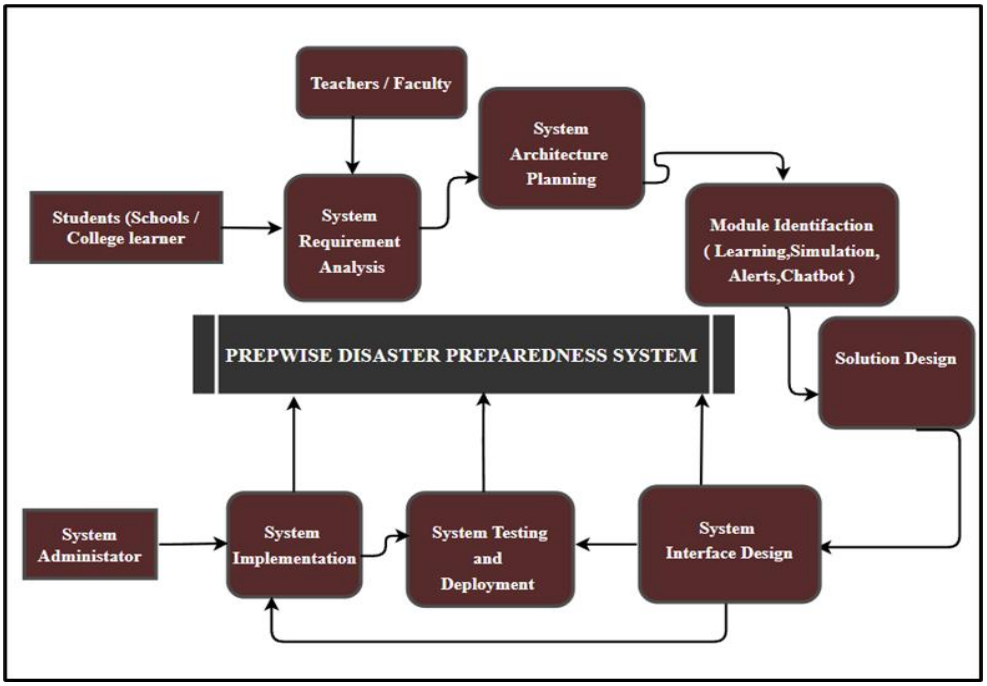


Figure 5.3 PREPWISE Development Workflow

Fig 5.3 indicates the PREPWISE disaster preparedness education platform domain model workflow. The figure displays the relationship between the system users and various development aspects of the platform. It starts with students and teachers who submit system requirements that rely on educational disaster preparedness requirements. The system requirement analysis stage involves the analysis of these requirements to determine the important functionalities needed in the platform. System architecture planning is done based on these requirements in order to define the general structure of the system. Once the architecture has been established, the module identification phase identifies the key functional parts of the architecture like learning modules, disaster simulations, alert systems and chatbot support. The solution design phase specifies the way that these modules will interact with the system. The stage of system interface design is concerned with the design of user interface where students, teachers, administrators interface with the platform. The system implementation phase will be done after the design stage, which builds the application modules and includes them to the PREPWISE disaster preparedness system. The system is then subject to test, and is deployed to give adequate functionality, reliability and accessibility in education facilities. The administrator is in charge of the system put in place and also makes sure that the platform is running smoothly. This is a systematic process that makes sure the disaster preparedness education system is well designed, implemented and rolled out to the schools and colleges.

Domain Model Specification

PREPWISE has domain model which depicts the relationship between the users and the system components. The PREPWISE system allows the user, be it a student or an administrator, to interact with it in different modules namely; learning, simulation, analytics, safety, chatbot and user profile.

All modules have dedicated functions like education, training, data analysis and support. The data storage system is linked to all the modules in that it handles user data, course content, and system information.

This model displays the relationship between different entities and the flow of data in the system.

It is also appropriate to the project because it provides a clear depiction of the organization and interaction of all key elements of the disaster preparedness platform.

5.9 Interaction with and Flow of Information

The system interaction and information flow presents the way in which the various components of the PREPWISE disaster preparedness education platform interact to deliver learning and safety awareness to schools and colleges. The platform functions based on a formal engagement between the user access interface with system services and information resources. When users go to the platform via the institutional portal, the actions they take, including choosing learning modules, initiating disaster simulations, checking preparedness guidance, or visiting the assistance feature provide requests in the system. The platform processing services take care of these requests and handle various functional modules of the system. The services access the needed information within the platform resources and produce the relevant outputs that can be educational content, results of the simulation, preparedness warnings, or guidance reactions. The system interface provides a channel to relay the processed information back to the user. This is a well-coordinated flow of information that facilitates the smooth liaison of system components and effective delivery of disaster preparedness education services. The integration architecture is useful in ensuring that the platform remains reliable in terms of performance, well structured data management, and interaction between the system modules that are developed with specific institutions of learning in mind.

5.10 Structure of System Deployments

The platform deployment structure is the way the PREPWISE disaster preparedness learning system is planned in all the parts of operation. The system is meant to aid the digital learning processes in the education institutions by linking learning interfaces, preparedness processing modules and information repositories. The system is connected to the users via digital access points, which include institutional learning portals or mobile interfaces. These entry points interlink with the preparedness platform that handles the simulations and emergency awareness exercise and preparedness guidance. The system maintains the preparedness resources and activity records in a central repository of information to facilitate the continuous learning and monitoring. Such a solution will keep the educational institutions capable of providing training on disaster preparedness effectively using the structured platform that is scalable. Furthermore, the deployment structure is designed to ensure high availability and secure access through cloud-based infrastructure, enabling multiple institutions to use the platform simultaneously without performance issues. The modular nature of the system allows easy integration of additional features such as real-time alert services and analytics dashboards. It also supports seamless communication between system components, ensuring efficient data flow and system responsiveness. This structured deployment enhances reliability, scalability, and adaptability of the platform in diverse educational environments.

Table 5.4 Preparation Requirement Assets of PREPWISE Platform

User Access	Devices that students and faculty (faculty) will use to access preparedness learning platform.
Learning Engine	Prepares modules, simulations and awareness exercises.
Content Repository	Stores disaster preparedness information, guidelines and learning sources.

Activity Monitoring	Monitors the user engagement and learning development in institutions.

5.11 FUNCTIONAL OVERVIEW

The most significant business capabilities of the PREPWISE disaster preparedness education platform are the functional activity structure. The platform can structure its functionalities in multiple synchronized activity groups that facilitate preparedness learning, system operation, and user engagement. These groups of activities deal with preparedness content provision, simulation activities, interaction support and monitoring of the systems. The functional groups play various roles in ensuring that an organized setting of disaster preparedness education in the learning institutions is achieved.

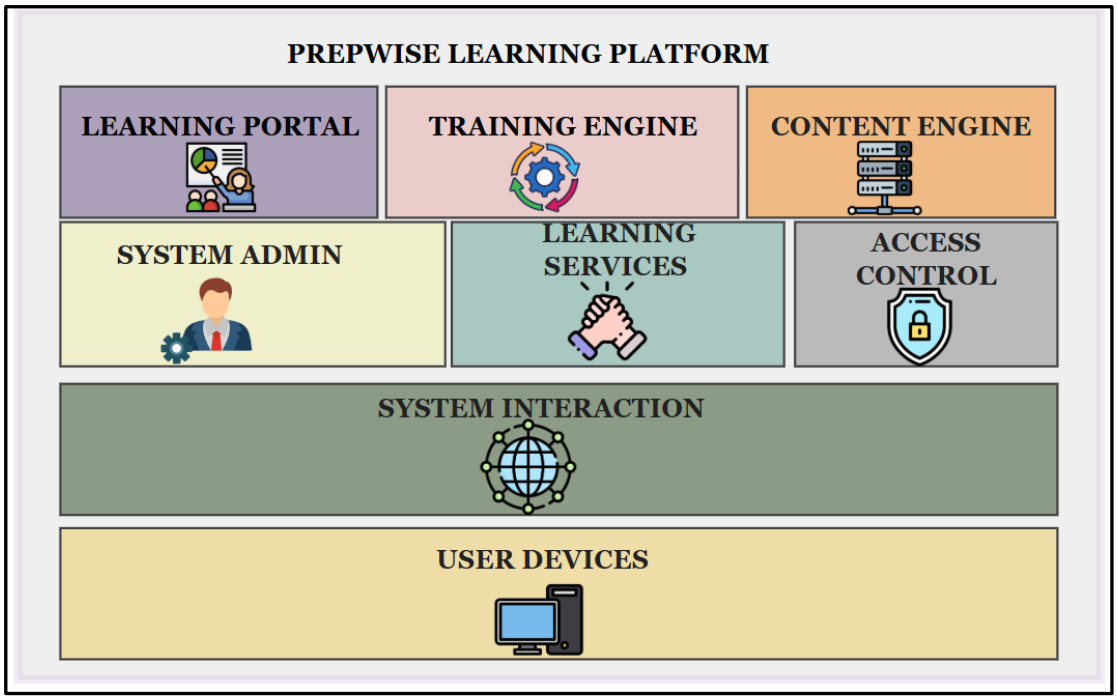


Figure 5.4 Learning System Architecture

The functionality of the PREPWISE disaster preparedness learning platform is depicted in **Fig 5.4** The most important layer is the primary platform on which the preparedness learning activities are provided by the training interface, simulation engine, and information repository. The second layer is the system core components including platform coordination, preparedness services and access protection that govern the system operations and interaction with the user. The system interaction layer provides

interconnection of the platform to the joined components via web interfaces and network services. The lower layer will be the user devices, which will be the computers and mobile devices to which the students and faculty will use to access the preparedness learning platform. These layers collaborate to promote successful disaster preparedness education in school and colleges.

5.12 Mapping Level of Deployment with Functional Blocks

The deployment view aids in outlining how the functional elements of the system are located onto the physical infrastructure where the application is being implemented. The proposed disaster preparedness platform has a deployment architecture composed of three major layers, the client layer, the application server layer, and the database layer. The client layer is the user interface on which the students, teachers, and administrators will interface with the system via a web browser. The platform allows the users to watch disaster courses, do simulations, take quizzes, and get disaster alerts. Application server layer handles the requests of the user and implements business logic of the system. It controls the activities of the system, including retrieving courses, performing simulations, answering chatbots, and sending alerts.

All the system data such as user profiles, disaster learning material, quiz result, and alert data is stored in the database layer. The application server will be used to relay to the database to obtain and update information whenever necessary.

This multidimensional deployment model provides scalability, efficient data processing, and safe data storage of system data.

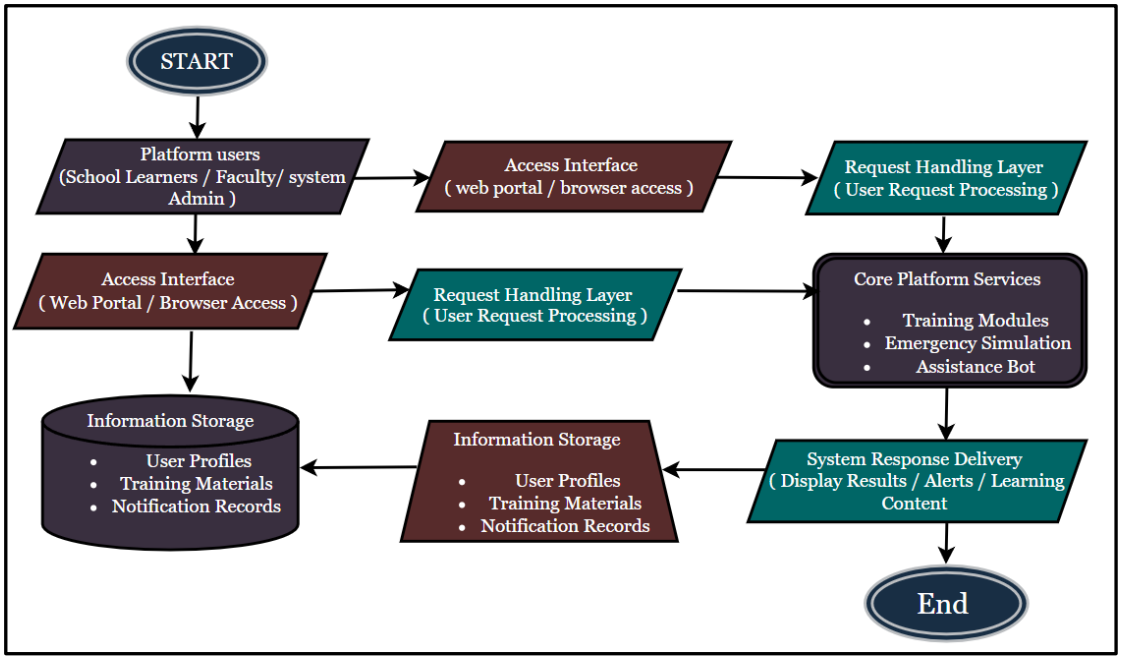


Figure 5.5 Component and Service Mapping

Fig 5.5 shows how the suggested PREPWISE Disaster Preparedness Education System will operate. The process starts by having the users of the platform like the students, faculty members, or system administrators use the system by accessing it via a web browser or portal interface. The interface of access is the point of interaction between users and the platform and place where they can make their requests to the learning modules, training material or simulation activity.

After a request is received, the request handling layer will take control of this request, handle user interactions, and send the request to the core platform services. Disaster preparedness training modules, an emergency simulation process, and an assistance chatbot, which assists the users with disaster-related guidance and information, are the essence of the core services. These services produce relevant answers in response to a request by the user.

The system also communicates with the information storage system which forms an important part of the system, which stores key data of the users like their user profiles, training material, as well as notification records. This storage allows learning content and user progress to be stored in the platform. Once the request is processed and the required data is found, the system presents the response to the user as an interface provided in the shape of learning material, warnings, or simulation results. The process is then completed when requested information or service is delivered successfully to the user.

5.13 Operational View

The operational view explains the workings of the system when it is being run in real-time. It encompasses communication mechanisms, service hosting options, and data storage elements that help in supporting the operation of the disaster preparedness platform.

In the proposed scheme, the web application is stored in an application server where all the user requests are processed and interactions among the parts of the system are handled. The web interface allows users to communicate with the system and pass their requests to the server via the normal web communication protocols. These requests are processed by the application server which obtains the corresponding information in the database server. Indicatively, upon a user visiting a disaster learning module, the server will retrieve the course material in the database and show it in the web interface. The chatbot service is an intelligent support element that works on the user queries and can provide the corresponding answers concerning the disaster preparedness and safety procedures. Likewise, the alert system constantly looks after disaster updates and provides the user with notifications in case of the need. The system communication is based on standard web protocols like on HTTP, as well as REST APIs so that the data exchange between various components can be efficiently conveyed.

Operational Overview of PREPWISE Disaster Preparedness Platform

- The PREPWISE Disaster Preparedness Platform is designed to provide disaster preparedness education and related services through an integrated system.
- The platform supports three main categories of users: students, teachers, and administrators.

1. Students can access learning modules, participate in simulations, and take quizzes.
2. Teachers can manage course content and monitor student progress.
3. Administrators handle user management, system configurations, and platform content.

- Users interact with the system through a client-side web interface using a web browser.
- The frontend is designed as a responsive web interface to ensure easy and user-friendly access across different devices.
- User requests are sent over the internet using HTTP/HTTPS protocols and RESTful APIs.
- The application server receives and processes all user requests and manages the core functionalities of the platform.
- The application server provides several key services, including:

1. Course and content management
2. Disaster simulation and quiz engines
3. Chatbot support for disaster-related queries
4. Alert and notification services

- These services ensure that users receive accurate learning materials and timely emergency guidance.
- A database server is used to store and manage important data such as course materials, user profiles, quiz results, and alert records.
- The platform integrates external services such as intelligent chatbot APIs for automated responses and real-time disaster data feeds providing updates on events like floods, cyclones, and heat waves.
- All components work together to form a unified system that enhances disaster preparedness education and improves awareness of disaster response.

5.14 Other Design Aspects

Process Specification

Process specification is where the steps of the system followed by the system as users interact with the various modules are outlined. As an illustration, a user can be identified by the system when the user logs into the system where the system will authenticate the user and then access the dashboard interface. On the same note, in case a user chooses a disaster learning module, the application accesses the corresponding content in the database and presents it on the web interface.

Specification of Information Model

The information model explains the way in which the data of the system is organized and stored in the database. The suggested system stores various forms of data such as user profiles, course contents related to disasters, simulation, quiz, and alerts. The data is done in structured or table database so that it can be stored and accessed effectively.

Service Specification

The service specification outlines the services that the system offers to aid the disaster preparedness education. Such services are disaster learning services, simulation services, chatbot assistance, alert notification services, and analytics services. Every service has a given functionality that plays a role in the entire functioning of the platform.

CHAPTER 6

SYSTEM IMPLEMENTATION AND SIMULATION

6.1 Deployment Environment

The PREPWISE system is a web-based disaster preparedness platform that does not require dedicated hardware components such as sensors, actuators, or microcontrollers. Hence, the project does not involve circuit diagrams or hardware interfacing for its operation. The system functions using standard computing devices such as smartphones, laptops, and desktop systems, which act as the primary user interfaces.All functional units of the system—including the dashboard, analytics, learning modules, simulations, safety features, and chatbot—are integrated through a centralized software architecture and require only a stable internet connection. Communication between these components is managed through backend processing and data storage mechanisms. A client–server architecture is followed, where the user device (client) sends requests to the server, and the server processes these requests and returns appropriate responses.This approach eliminates the need for specialized hardware setups and ensures that the system is easily accessible to users from diverse backgrounds.

6.2 Software Development Tools

The construction of the PREPWISE system entails a range of software applications that facilitate the entire development cycle comprising of coding, testing, debugging, version control as well as deployment. These tools are useful in enhancing productivity, quality of codes, and efficient cooperation in a team.

Code Editor / Development Environment

The application code is written, edited and managed with the help of a code editor. It gives the features like syntax highlighting, auto completion, debugging support and integration of extensions that make it more efficient and fast in development.

Configuration:

Code editor is installed and set with extensions required to do web development. The project files are put together in folders which are arranged into frontend and backend

folders. There is a good practice in file naming and modular coding to make it maintainable.

Version Control System

A version control system is adopted to monitor the changes in the code and take care of various versions of the project. It enables several team members to be working on the project at the same time without conflicting.

Configuration:

A common repository is established where all files of the project are stored. Regular changes are made, and they are synchronized to ensure that there is consistency. Version control is also used to recover a past version of the code when necessary.

Web Browser

The application is tested and executed using a web browser. It allows developers to have a look at the user interface and test various system functionalities.

Configuration:

The application is deployed on a local or a server and features like a login, the dashboard navigation, learning module, simulations, and chatbot responses are tested in the browser. The browser also has developer tools that are utilized in debugging and performance analysis.

API Testing Tool

The communication between the frontend and the backend components is tested using API testing tools. They assist in making sure data is relayed and fetched properly among other modules of the system.

Configuration:

The testing is done by the API using requests and responses. This will guarantee the correct operation of all the modules like authentication, data retrieval, and chatbot interaction.

Deployment Platform

The application is operated on a deployment platform so that it can be accessed by the users via the internet. It also makes sure that the system is able to be accessed at other places without installation.

Configuration:

The application is implemented on the cloud or server. Proper configurations like server, routing and database connection are put in place to achieve a smooth running of the system.

Testing Tools

The performance, reliability and functionality of the system are tested with the help of testing tools. They assist in diagnosing the mistakes and make sure that each of the modules functions as anticipated.

Configuration:

The different features that are required in the design and implementation of the test cases involve user authentication, entry to modules, running the simulator and communication with chatbots. The system is subjected to varying conditions so as to test the stability and accuracy.

6.3 Software code

Manifest.json:

```
{
"name": "PrepWise — Disaster Preparedness",
"short_name": "PrepWise",
"description": "Real-time disaster alerts, emergency guides, and safety planning",
"start_url": "/",
"display": "standalone",
"background_color": "#ffffff",
"theme_color": "#185FA5",
"icons": [
  {
    "src": "/icons/icon-192.png",
    "sizes": "192x192",
    "type": "image/png"
  },
  {
    "src": "/icons/icon-512.png",
    "sizes": "512x512",
    "type": "image/png"
  }
```

```
        ]
      }
```

Service-worker.js:

```
const CACHE_NAME = "prepwise-cache-v1";
const urlsToCache = [
  "/",
  "/offline.html"
];

self.addEventListener("install", (event) => {
  event.waitUntil(
    caches.open(CACHE_NAME).then((cache) => {
      return cache.addAll(urlsToCache);
    })
  );
});
```

Basic Express Backend (Node.js):

```
self.addEventListener("fetch", (event) => {
  event.respondWith(
    fetch(event.request).catch(() => caches.match("/offline.html"))
  );
});
const express = require("express");
const app = express();

app.use(express.json());

app.get("/alerts", (req, res) => {
  res.json([
    { type: "Flood", location: "Bangalore", level: "High" },
    { type: "Earthquake", location: "Delhi", level: "Medium" }
  ]);
});

app.listen(3000, () => {
```

```
  console.log("Server running on port 3000");
});
```

6.4 Simulation

The PREPWISE system incorporates software-based simulation, rather than hardware simulation. The simulation aspect enables customers to undergo the virtual disaster situation and know the correct way to respond.

Simulation Approach

- The application simulates the effect of the disaster like floods.
- Simulations require users to interact with the system and make some decisions.
- The system gives feedback in response to the activities by the users.

Tools Used for Simulation

- Simulation based on frontend logic through the web.
- Test environment and execution environment Browser environment.
- Simulation logic Backend processing of simulation logic.

Description

In PREPWISE, simulation assists users to get familiar with the issue of a disaster in the real life without any physical danger. It enhances preparedness as it gives users the opportunity to train on decision making during emergencies. The system provides a correct representation of scenarios and gives guidance depending on the interaction of users.

Suitability for the Project

The software based simulation is appropriate in the project because it does not require physical hardware, and it can be accessed by many users. It is also scalable and interactive in real-time and therefore useful in disaster preparedness education.

CHAPTER 7

EVALUATION AND RESULTS

7.1 System Evaluation Overview

The Disaster Preparedness and Response Education System (PREPWISE) was evaluated to investigate the functionality, usability and performance of the system. The system is developed with several modules such as disaster learning courses, simulation exercises, dashboard analytics, emergency notification services, and chatbot support that can be integrated to create a holistic disaster preparedness learning system. Different test cases were modelled and carried out to ensure that each of the modules is working in line with the requirement of the system and is giving correct outputs.

The testing procedure was mostly aimed at testing the communication between users and the system. Various activities of users like accessing courses on disaster preparedness, taking part in simulation exercises, as well as engaging the chatbot to get safety guidance and track disaster alerts were experimented with. Each of the functions was tested according to the requirement that the platform should react properly to the inputs of the users and provide the desired output without failures.

Besides the functional testing, the usability of the platform was also monitored to see whether the platform was easy to navigate and whether users could access the available learning resources easily. System dashboard was also tested based on displaying progress information, analytics and alerts in a well organized way. These tests assisted in confirming the existence of user-friendly environment within the platform, which supports both emergency awareness and learning.

The outcomes of the evaluation indicate that the designed system is efficient and addresses the planned goals of disaster preparedness education. The platform is effective and can handle user requests by accessing pertinent information in the database and responding accordingly in all the modules. Altogether, the PREPWISE system is an interactive and efficient learning platform which can assist students and educational institutions in their awareness, knowledge, and preparedness to react to the natural disaster.

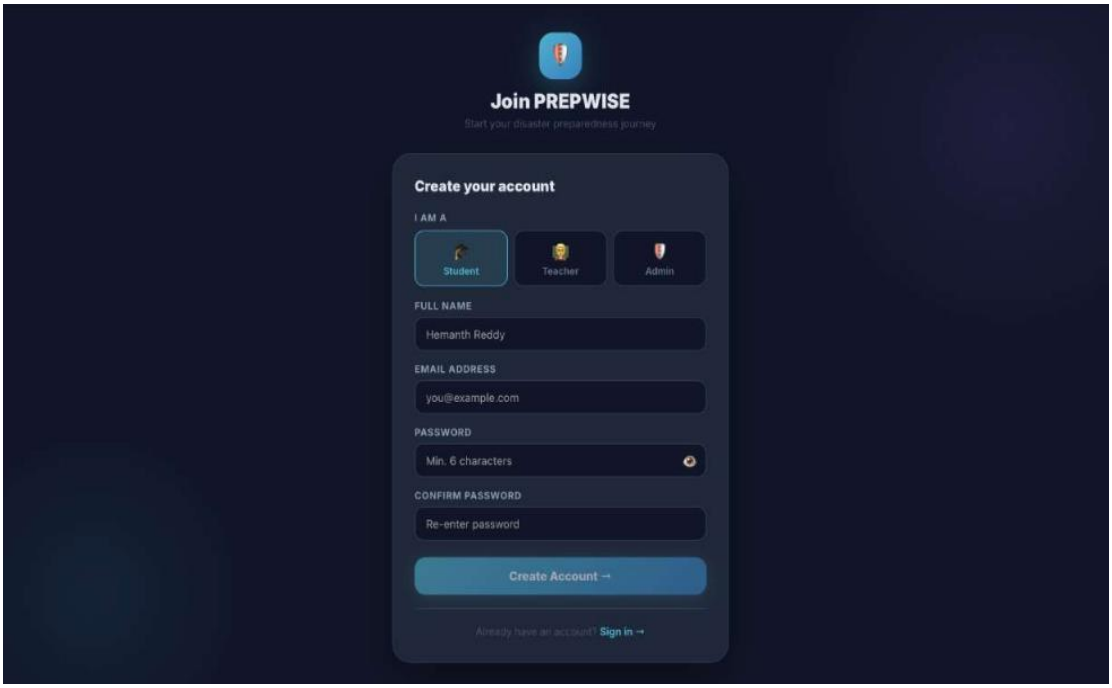


Figure 7.1 User Registration Interface

The figure above shows a user registration interface of the PREPWISE system, which was created to enable new users to create an account easily. The interface offers a role-based selectivity that allows the users to sign up as a Student, Teacher or Admin, and this information ensures proper role-specific accessibility in the system.

The shape gathers necessary user information including full name, email address and password. It has a confirmation password field to reduce the occurrence of input error and increase data precision. The password field has simple validation standards, including minimum required number of characters, to enhance security.

It has a clean and modern user interface design and a centralized form layout that is user friendly and easily readable. The module is essential in authenticating the users and allowing them to have customized access to disaster preparedness resources and functionalities in the platform.

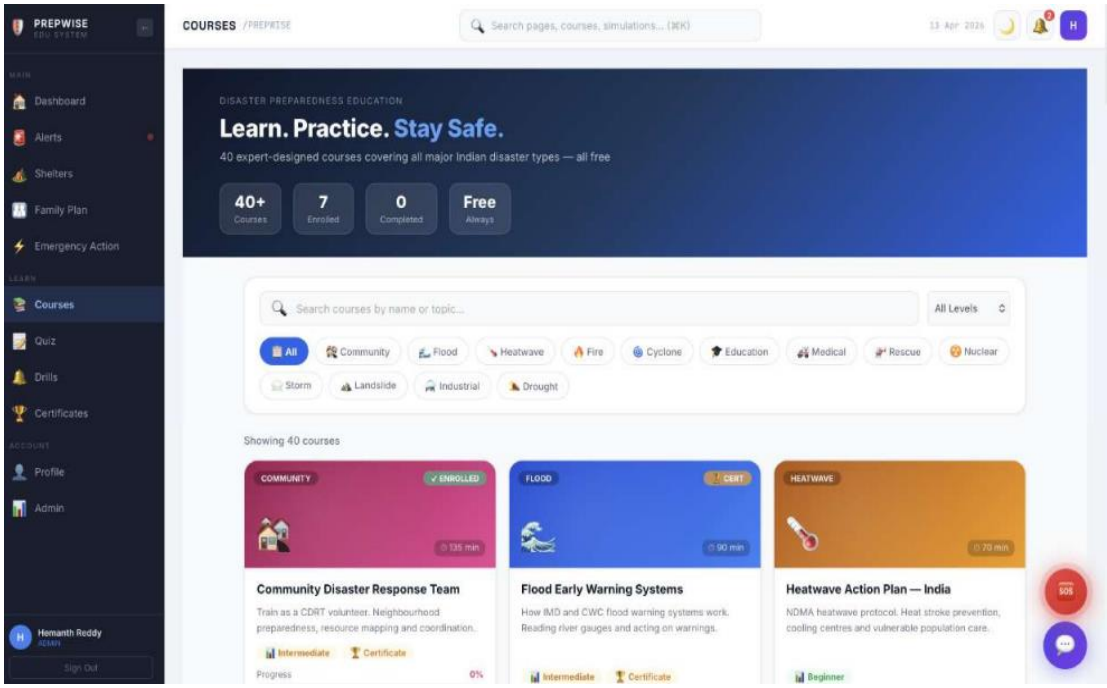


Figure 7.2 Learning Modules Page

This screen offers some of the disaster preparedness courses that users can take. It enables a filtering by categories such as flood, fire and cyclone. The page assists the users to study and enhance their emergency response

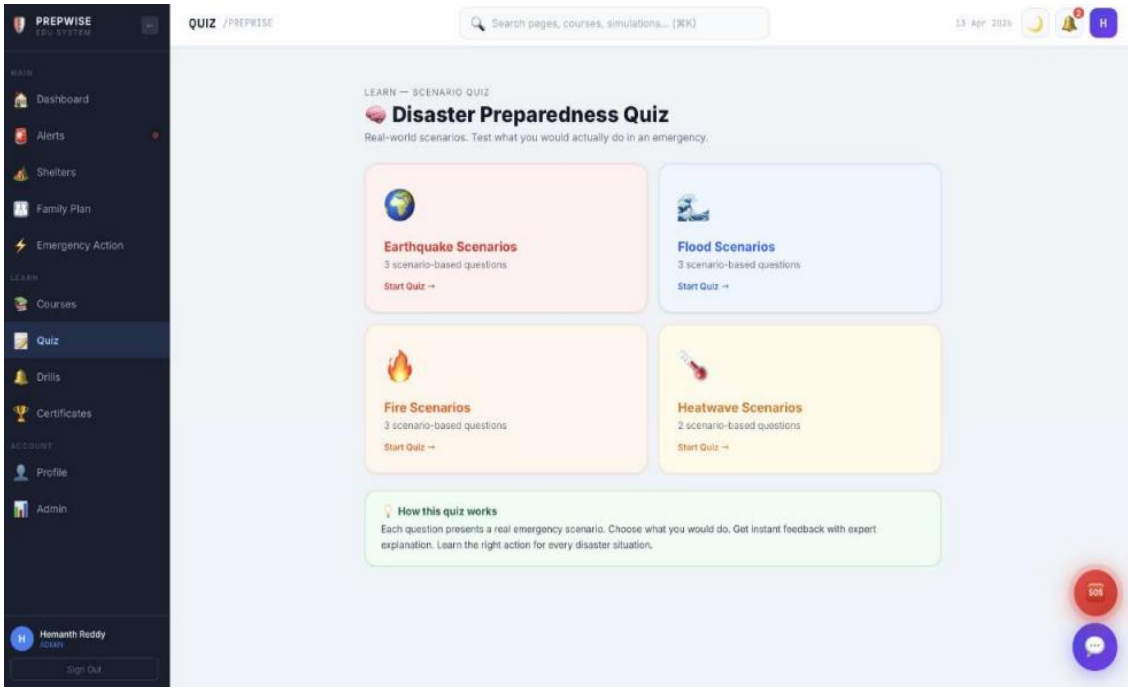


Figure 7.3 Quiz and Scenario Tests

This screen depicts the disaster preparedness quiz section with various scenario based categories such as earthquake, flood, fire, and heatwave. Users have the opportunity to

take quizzes to check their knowledge. It assists in enhancing decision-making under emergency occasions

7.2 Functional Module Evaluation

The evaluation of the functional module of the application will be performed according to the following criteria:7.2 Functional Module Evaluation The evaluation of the functional module of the application will be analysed based on the following criteria:

The system developed is comprised of various modules that are intended on enhancing the disaster awareness and preparedness of users. Each of the modules was tried individually to see whether that particular module does the job it should do.

The key modules analysed in the system are:

- The authentication of users and management of their profile.
- Disaster learning courses
- Disaster simulations on the Internet.
- Quiz and training modules
- Disaster monitoring panel.
- Emergency alert system
- Chatbot assistance

Predefined input scenarios were used to test each of the modules and analyse the resulting system outputs. The testing ensured that system modules are operating properly and provide the correct results.

Table 7.1 Testing Results of System Module

Module	Input action	System output	Status
Login system	User enter credentials	Successful login	Passed
Dashboard	Access analytics	Correctly displayed data	Passed
Courses	Disaster course selected	Learning module loads	Passed

Exercise	Disaster drill simulation begins	Disaster drill simulation begins	Passed
Chatbot	Ask safety question	Response generated	Passed
Alert System	Disaster warning activated	Notification shown	Passed

7.3 System Performance Analysis

The system performance analysis does assess the level of performance of the system as per its intended purpose. The system performance was measured in terms of analysing the response time and efficiency of various modules. The findings show that the system is effective in receiving user requests and providing disaster preparedness data. The authentication system of the system is performed within a short time, as the login module authenticates the users and grants them a secure access to the system. Dash loads system analytics and statistics without a lot of latency. The disaster learning modules enable the users to interact with the educational content without any problems, whereas the simulation module gives them the opportunity to learn through interactive disaster situations.

The chatbot is also fast in responding to user queries and giving them the right instructions on disaster safety. Generally, the system has an excellent performance in all the modules.

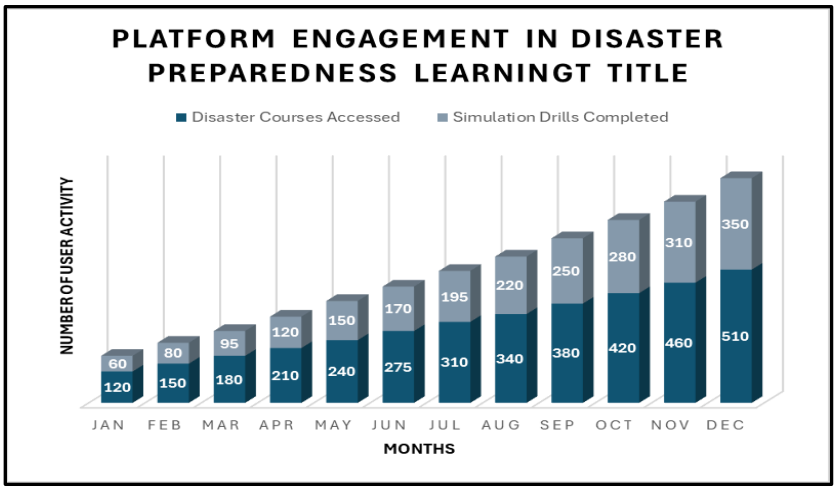


Figure 7.4 Platform Usage Trends

In the Above graph **Fig.7.4** , It can be seen that the number of users accessing disaster courses and going through simulation drills is gradually rising and thus this is an indication of better interaction with the system. The results of the chatbot interaction will be presented in this section

7.4 Chatbot Interaction Results

The chatbot module will offer automated support to the users by addressing disaster-related inquiries and dispensing safety measures. This will enable the students to get disaster preparedness information fast without the need to search the entire system.

In the process of testing, several questions were inputted to the chatbot on matters concerning disaster response procedures. The chatbot managed to handle such queries and came up with pertinent answers depending on the disaster knowledge base.

The findings suggest that the chatbot would make the disaster preparedness information more easily accessible and the overall user experience of the platform more meaningful.

Example input:

User: Flood Safety Guide

Output:

Chatbot: Flood Safety Guide

The floods may be disastrous and fatal.

It's essential to take necessary precautions to ensure your safety.

The following is a detailed guide on how you can prepare and react to floods:

Before a Flood

- Create a flood emergency kit with essential items such as:
- Water (at least 3 days' supply)
- Non-perishable food

First aid kit

- Flashlights and batteries
- Battery-powered radio
- Whistle to signal for help
- Stay informed about weather conditions and flood warnings in your area.
- Know your evacuation route and have a plan in place.

During a Flood

- Stay indoors and away from floodwaters.
- Avoid walking or driving through floodwaters, as they can be contaminated with sewage, chemicals, and other hazards.
- Keep phone lines open for emergency calls only.
- Unplug appliances and electronics to prevent damage from power surges.
- Stay tuned to local news and weather reports for updates.

Evacuation

- Follow evacuation orders from local authorities.
- Take your emergency kit with you.
- Stay on designated evacuation routes.
- Avoid flooded areas and take alternative routes if possible.
- After a Flood
- Wait for official clearance before returning to your home.
- Be cautious when entering your home, as floodwaters can cause structural damage and hide hazards.
- Check for damage and take photos for insurance purposes.
- Discard contaminated items, such as food and water, that have come into contact with floodwaters.
- Seek medical attention if you've been exposed to floodwaters or have injuries.

Important Contacts

- National Emergency: 112
- NDRF (National Disaster Response Force): 1078
- State Disaster Management: 1070
- Remember, flood safety is everyone's responsibility. Stay informed, be prepared, and follow evacuation orders to stay safe during a flood.

7.5 Analysis of System Response Time

The system modules response time was measured to test the efficiency and performance of the PREPWISE platform in general. The response time is the time of the system to accept user input, process the request and produce the relevant output. According to the

test results, it was seen that majority of the modules involved in the system take a very short period of time to respond, which ensures that there is no interruption in interaction between the user. Frontend rendering optimization and effective data processing mechanisms resulted in fast response times in modules like login, registration, dashboard loading, and displaying alerts.

Nevertheless, response time was relatively low with the simulation module. This is mainly because it entails loading of interactive data, real-time situations and dynamic data processing, which need more computing resources. Nonetheless, the delay is still acceptable and does not have a serious impact on user experience.

In addition, the performance of the system does not reduce when several functionalities are used at the same time. It means that the platform can accommodate a moderate user load without any significant drop in performance and responsiveness.

On the whole, the analysis of response time shows that the PREPWISE system is able to perform on the tasks of an educational application. The system is also used to deliver the information in a timely manner, which is especially important in the case of disaster preparedness when it is necessary to access information promptly and obtain guidance and alerts.

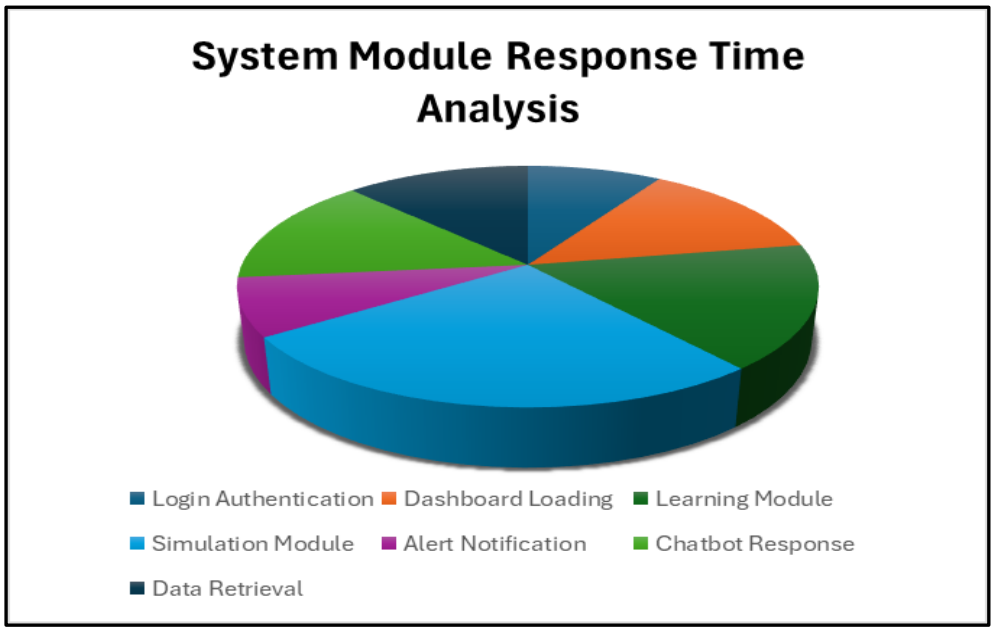


Figure 7.5 Response Time Comparison

The pie chart **Fig 7.5** shows the response time of various modules in PREPWISE platform. The modules of the system that exhibit better response time are the login and

the alert notification modules, and the process time of simulation module is a little higher because it performs the interactive disaster scenarios. In spite of this deviation, all the modules have got their work at the tolerable limit of response, which guarantees easy performance of the system in learning disaster preparedness.

7.6 Results of Tests of System Modules

The functional testing of PREPWISE Disaster preparedness platform was conducted to determine whether every module works as per the anticipated system requirement. Scenarios were run to test the behaviour of the login system, learning modules, disaster simulations, alert notification, and the chatbot behaviour. The findings assure that all the modules are functional and deliver the desired outputs when the system is executed.

Table 7.2 Evaluation of the performance of the system

Order	**Mean Response Time(sec)**	**Performance Level**
Login authentication	0.8	Very fast
Dashboard loading	1.2	Fast
Learning module	1.5	Fast
Simulation module	2.4	moderate
notification of alerts	0.7	Very fast
Chatbot response	1.3	Fast

7.7 Observations and Insights

- The analysis findings provide a number of key insights into the work of the disaster preparedness system.
- The dashboard analytics assists the administrators with tracking disaster preparedness activities suitably.
- The disaster learning modules enhance awareness of the students on disaster response techniques.

- The simulation module will contribute to a better appreciation of the safety operations in the case of emergency situations.
- The chatbot enables fast access to disaster safety and enhances the interaction of the user with the system.
- The alerts system facilitates timely declaration of the disaster warnings to the users.

Comprehensively, as the evaluation results indicate, the developed platform manages to combine disaster education, simulations, and monitoring capabilities into a single platform that can be useful in enhancing the disaster preparedness in educational institutions.

7.8 Future Improvements

The system offers several opportunities for future enhancements to improve its functionality and user experience. One of the major improvements includes the integration of advanced artificial intelligence techniques to provide personalized learning recommendations based on user performance. This will help users focus on areas where they need more improvement. The development of a dedicated mobile application can increase accessibility and allow users to access learning resources anytime and anywhere. This is especially useful in emergency situations where quick access to information is required. Another potential enhancement is the integration with government disaster management systems to provide real-time alerts and official emergency updates. This will improve the reliability and usefulness of the platform during actual disaster situations.

7.9 Results

INPUT

CODE

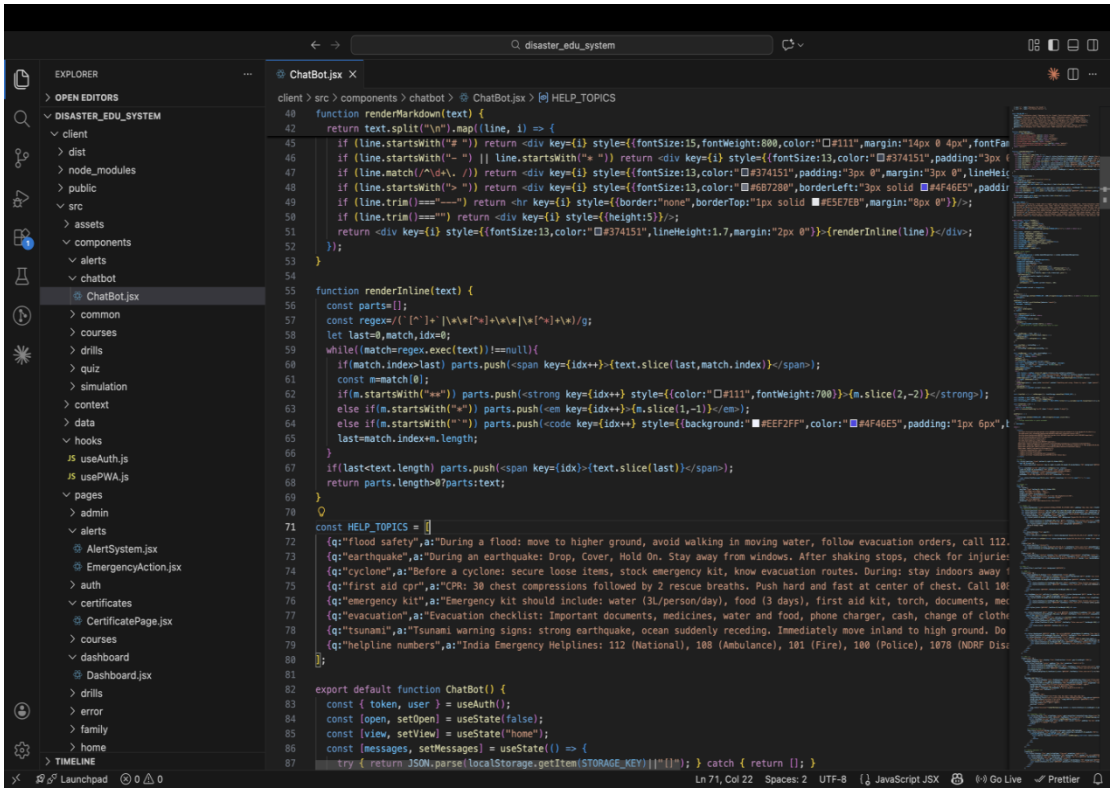


Figure 7.6 Chatbot Implementation Code

OUTPUT

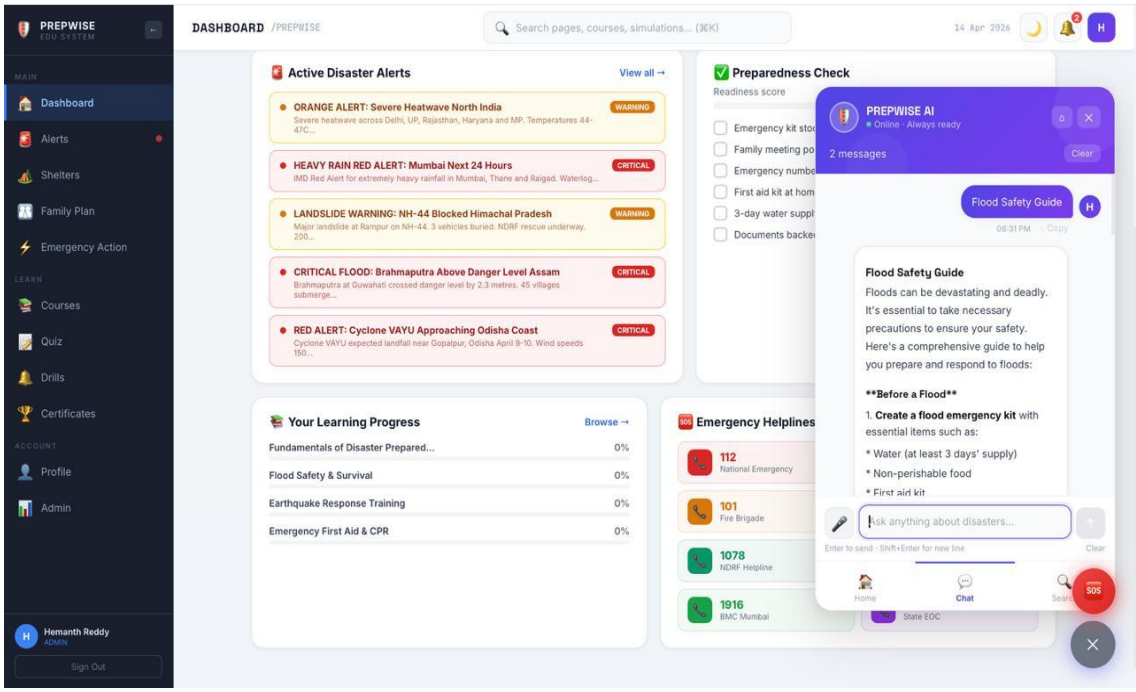


Figure 7.7 AI Chatbot for Disaster Support

This screen displays the incorporated chatbot that is helpful when it comes to questions on disasters and safety guidelines. It offers prompt interactive feedback and answers to guide users to do the right thing. The chatbot enhances the availability of emergency information

CHAPTER 8

ETHICS, SAFETY AND SUSTAINABILITY

The evolution of digital systems which will facilitate preparedness of disaster in institutions should be in the light of wider social, legal, and moral liabilities. A system aimed at teaching students on emergency response should be reliable, fair and handle data with care and also benefit the society. The Disaster Preparedness and Response Education System in Schools and Colleges is focused on enhancing awareness and preparedness with the help of digital learning modules, simulated drills, real-time notifications, and chatbot assistant.

In this chapter, the interaction of the developed system with the social structures, its adherence to legal frameworks, compliance with ethical standards, sustainability, and the safety of the users are discussed.

8.1 Society and Community Good

The shift to digital channels in disaster preparedness has a great impact on the society with a better awareness, communication, and emergency preparedness. Schools can be valuable in creating disaster resilient communities, and learning environments should include preparedness education that will help avert panics and confusion in times of crisis. The suggested system is useful to society in a number of ways. To begin with, it increases the disaster awareness levels among the students and staff members through offering systematic educational programs on earthquakes, floods, fire and other natural disasters. Second, virtual exercises enable users to train on response strategies in a simulated setting without the logistical burden of conducting a large-scale physical exercise. Third, geographic-based alerts are used to equip institutions with earthquake responses associated with their region.

Socially, the platform helps in the development of preparedness culture. Students educated by this system can impart knowledge to their communities and families, and thus increase the reach beyond the schools.

Nonetheless, there should be technological solutions to challenges. The lack of an equal access to the digital resources can restrict the participation of the institutions located in the rural or less economically advantaged areas. To minimise this digital divide, the

system will run on a basic web based browsers and mobile gadgets with few technical specifications.

8.2 Data Regulation and Compliance

Electronic systems that process the data of users have to adhere to appropriate legislation and laws. The Disaster Preparedness Education System holds user profiles, drill participation information, and emergency contact information. Thus, the appropriate data governance systems are in place to fulfil the data protection regulations.

In India, the Digital Personal Data Protection Act (DPDP Act), 2023 contains the guidelines of responsible treatment concerning the individual information. The major principles of the system include the use of consent to collect data, restricted use of the data, and safe storage procedures.

It is the responsibility of institutions that use the platform to ensure that the information about students is gathered with due diligence. The site also allows administrators to control data access privileges and to block improper use of sensitive data.

The other aspect that is an essential consideration of law is the accuracy of the information presented in the system by the emergency. The information about the disaster alerts has to be correct and confirmed, preferably to be the one of the official authorities like disaster management agencies or the government notification systems.

The inability to meet the legal requirements might result in the abuse of the data, the absence of the trust, and legal risks facing institutions using the platform. Thus, accountability and compliance are encompassed with secure authentication, encrypted methods and frequent monitoring.

The topic of social responsibility in disaster education technology is also covered in ethical responsibility (Lynch, 2018).

There are high ethical obligations linked with technology having an impact on human safety. The developers should make sure that the system is able to promote fairness, transparency, and reliability.

The public good is one ethical principle that was used in developing this system. The main aim of the platform is to save lives by enhancing disaster preparedness. Thus,

design choices are made in such a way that they are accessible, easy to use and understand by students and staff.

The other ethical issue is the veracity of educational material. The misinterpreted knowledge concerning emergency protocols might result in unsafe outcomes in case of any calamities. So as to avoid this problem, the educational content provided in the system is founded on proven disaster management rules.

Moral issues are also connected with the presence of a chatbot helper. During the emergency situation, the chatbot will assist in general advice, however, it should make it clear that it is not to substitute the official emergency services. This helps avoid excessive dependence on the use of automated responses in situations of crisis.

Also, the system does not have addictive or distracting features that may diminish the role of education. The gamification features are applied to stimulate the interest to the preparedness learning modules instead of establishing an unwarranted dependency.

8.3 Environmental Sustainability of the Digital Platform

The factor of sustainability is becoming a critical aspect of software development. Even though no physical waste is generated through digital platforms as it is created through manufacturing, it consumes computation resources and energy.

The Disaster Preparedness Education System is developed with the sustainability in mind through the adoption of effective development practices. The use of clouds simplifies the processes of physical infrastructure in every institution. Shared servers conveniently enable more than one school and college to use the platform as they reduce on hardware. The system also enhances sustainability as it will minimize frequent physical training that necessitates use of printed materials, transportations among other logistics resources. The platform can be used to undertake virtual exercises on emergency response, which can help an institution to rehearse emergency responses without causing environmental damage.

Moreover, the platform supports the long-term sustainability through educating students concerning disaster preparedness and environmental consciousness. Knowledge of natural hazards and risks related to climatic conditions can assist young people in responsible attitudes to environmental protection.

8.4 System Safety and Cybersecurity

The safety of any system intended to provide disaster response is a very important feature. The security and integrity of platform should be guaranteed to ensure that users are correctly guided in case of an emergency.

The system has a number of safety features. Authentication checks are used to store control over authorized users like students, teachers and administrators. Strict login systems ensure that user data cannot be accessed by an unauthorized user. Also, data interception is avoided via the encryption of communication channels.

The security measures related to cybersecurity are crucial since the malicious attacks might interfere with the emergency communications. Such risks are minimized by the use of regular security updates, vulnerability testing, and monitoring mechanisms.

The chatbot assistant will be developed with safety limitations. It gives overall instructions that follow the established knowledge of responding to the emergency but does not produce crucial instructions that can result in harm. When faced with actual emergencies, the chatbot prompts a user to call the official emergency services or adhere to the safety guidelines put in place by the institutions.

These security measures have made the platform reliable, secure, and trustworthy in the normal learning processes and in emergencies.

8.5 Metrics of Preparedness Evaluation

To determine the effectiveness of the platform, the level of readiness was measured depending on the participation of students on learning modules and virtual training. Preparedness score is the ratio of students who have successfully gone through disaster awareness modules and took part in the simulation-based exercises offered by the site. This measure is used to assess the level of engagement of the users with the preparedness training content.

Besides participation, completion rates of modules and performance during simulation assessments are also considered in the system to identify the general level of preparedness. The higher the participation and completion rates the greater the awareness and the level of institutional readiness. Such measures assist the educational facilities to track the efficiency of disaster preparedness training and define the areas where more training or awareness programs could be necessitated. Comprehensively, a

better preparedness score will indicate better student ability to act adequately in case of emergency

Table 8.1 Preparedness Score Based on Module Participation

Institution	**Total students**	**Students completed training**	**Participation rate (%)**	**Preparedness score**
College A	200	170	(170/200) x 100=85%	High
College B	150	105	(105/150) x 100=70%	Moderate
School C	120	96	(96/120) x 100=80%	High
School D	180	108	(108/180) x 100=60%	Average

The preparedness score will be a proportion of students who have successfully attended disaster awareness modules and undergone virtual drills. An increase in the participation rates depicts an improved institutional preparedness and enhanced preparedness in case of an emergency situation.

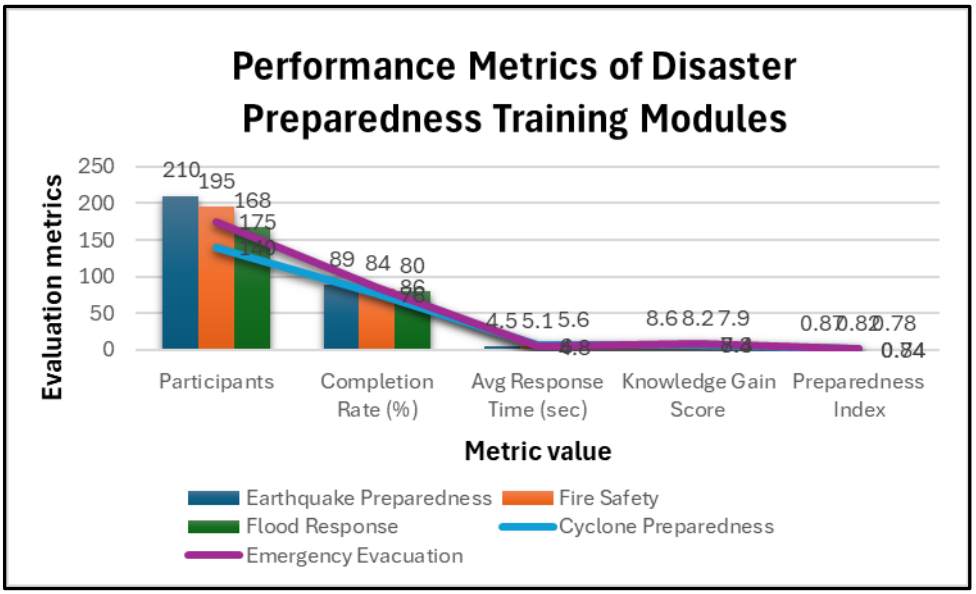


Figure 8.1 Comparative Disaster Preparedness Training Module Analysis

The chart reflects comparative analysis of the various disaster preparedness training modules applied in the system under the following, Earthquake Preparedness, Fire

Safety, Flood Response, Cyclone Preparedness, and Emergency Evacuation training. The effectiveness of each module was determined by several performance indicators that included the number of participants, the rate of modules completion, mean response time, knowledge gain score, and preparedness index, according to which the Earthquake Preparedness training was the most successful one with the highest number of participants (210) with better perceived importance of seismic safety and high knowledge. Regarding the completion rates, Emergency Evacuation and Earthquake Preparedness modules were associated with a higher level of engagement, which means that students actively engaged in reading the training material. The average response time indicator reflects the duration of time required of the participants to correctly provide answers to simulated emergency incidents during virtual drills. Faster response time indicates enhanced knowledge of safety measures. Also, the knowledge gain score, which assesses the amount of disaster awareness following the training modules, is the final measure it is a composite of the completion rate, response efficiency, and knowledge gain into a normalized measure of disaster preparedness. The findings indicate that preparedness levels are greatly positively correlated with the modules, which indicate the efficiency of the digital disaster education system in raising awareness and response capacity in case of an emergency among the students.

CHAPTER 9

CONCLUSION AND FUTURE WORK

9.1 Conclusion

The Disaster Preparedness and Response Education System of Schools and Colleges was created to help with the increasing demands of systematic disaster preparedness education in schools and colleges. Conduction of regular disaster management training in many schools and colleges is a challenge because of the limited resource base, absence of coordination and poor awareness of the students and staff. This can cause people to fail to react properly in an emergency like earthquakes, floods, fires, or cyclones. The key goal of the project was to design and introduce a digital platform which will offer convenient disaster education in the form of interactive learning modules, simulated drills, and real-time guidance tools. The designed system incorporates several functional components that will help to increase the rates of disaster awareness and preparedness. These modules are disaster awareness learning, disaster simulation games, emergency contact directories, preparedness monitoring dashboard, and a chatbot-based assistance. With this functionality, the platform allows students and staff to get to know the necessary procedures during disaster responses and rehearse emergency response actions in a systematic and interactive way. The online solution embraced in this project goes a long way towards increasing the effectiveness of disaster preparedness training than manual drills. Through analytical dashboards, students have access to learning materials, get to experience simulated disaster scenarios, and assess their levels of preparedness. The inclusion of gamification as the learning features also make the students more involved and make them eager to engage in the disaster awareness programs. This work is based on a project aimed at determining outcomes and accomplishments.

9.2 Project Outcomes and Achievements

The introduction of Disaster Preparedness Education System has led to a number of significant outcomes that can help to enhance safety awareness levels in learning institutions. The system is effective in achieving the goals set at the beginning of the project by availing an effective digital solution to disaster preparedness training.

The key results of the project include:

Better disaster awareness: Students develop a more in-depth comprehension of disaster-related safety measures of earthquakes, floods, fires, and other emergencies.

Increased engagement of the students: Virtual exercises and interactive modules will motivate them to participate in disaster preparedness training.

Effective training management: The administrative dashboard makes it simple to conduct and track training programs on disaster management in educational institutions.

Safety information at their fingertips: The in-built chatbot enables users to receive advice on how to proceed in the event of an emergency and safety measures.

Performance observation: Readiness measures, including participation rates, rates of knowledge gain, and response rates assist in determining the success of training programs.

According to the system evaluation results that were provided in the last chapter, there are positive performance indicators that comprise high levels of participation in the training and increased levels of preparedness amongst the students. As a result, the discussion of training modules such as earthquake preparedness, fire safety training, flood response training, cyclone preparedness training, and emergency evacuation training reveals that digital learning environments can be efficient in increasing the knowledge retention and emergency response capabilities. Moreover, the chatbot feature will increase the usability of the system by responding with automated responses to disasters-related queries. The users are able to receive information on safety precautions, evacuation routes, and emergency contacts very fast. On the whole, it is possible to say that the effective development and deployment of this system prove that digital platforms can be one of the fundamental components in enhancing disaster preparedness education and establishing less risky learning conditions.

9.3 Future Work

Even though the offered system has managed to reach the main goals, there are some areas to be improved and extended. The technological innovations in artificial intelligence, mobile computing, and immersive simulation technologies could be employed to make the platform functional and effective in the future and further

development of the system can be done to make the platform functional in the large-scale institutional use, integration with the official disaster management systems, and the enhancement of user experience. With the introduction of innovative technologies and other functionalities, the system can become a universal disaster preparedness ecosystem and serve not only institutions of education but also the emergency management departments.

9.4 Possible Improvements and Recommendations

Some upgrades can be taken into account in the next generation of the Disaster Preparedness Education System to make it more effective and efficient.

Technological Enhancements

Connection of real-time disaster warning systems based on data provided by the meteorological departments or state disaster management agencies.

Optimization of the AI chatbot system with the help of sophisticated Natural Language Processing (NLP) methods to have more precise responses and be able to comprehend the context.

Creation of a mobile-based application that will enable students and teachers to access disaster preparedness information using their smarter phones and tablets.

Scalability through the use of cloud to have more than one institution using the system at the same time.

Simulation and Training Enhances.

Integration of Augmented Reality (AR) and Virtual Reality (VR) technology to design immersive simulation environments of disaster simulation.

Creation of interactive scenario based training programs that mimic disaster scenarios in the real world.

Implementation of gamification elements like achievement badges and progress tracking to improve the motivation of students.

Expansion Opportunities in the System.

Liaison with local disaster management to be able to offer confirmed safety measures and emergency response plans.

The platform should be expanded to cover community awareness programs such that parents and local communities should also engage in the disaster preparedness education process.

Application of better data analytics to the performance of the training and where the

- levels of preparedness should be improved.
- Research and Development Opportunities.
- Predictive disaster risk model development on past data and machine learning.
- The geographical information system (GIS) to give location-based disaster awareness and assessment of risk.

To assess the long-term effects of digital disaster education on the preparedness levels of students, conducting massive user studies would be appropriate.

To sum up, the Disaster Preparedness and Response Education System of Schools and Colleges shows how online technologies can play a crucial role in enhancing disaster awareness and emergency preparedness in schools and colleges. The system offers an all-encompassing answer to the disaster education issue by combining educational technologies with disaster management strategies to make the learning environment safer and more resilient. The platform will become a strong disaster preparedness ecosystem that helps educational establishments, communities, and disaster management authorities to create the culture of safety and resilience with a further improvement and incorporation of innovative technologies.

REFERENCES

[1] Sharma and R. Singh, Mobile learning to rural education, IEEE Access, vol. 9, pp. 12345–12352, 2021.

[2] P. Verma, S. Gupta, and R. Mehta, Opportunities and challenges in cloud-based education systems, IEEE Trans. Learning Technologies, vol. 13, no. 2, pp. 112120, 2019.

[3] S. Pozdniakov, R. Martinez-Maldonado, S. Singh, P. Chen, D. Richardson, T. Olivier, and D. Gasevic, Question-driven learning analytics: Designing a teacher dashboard in an online breakout room in Proc. IEEE Int. Conf. Advanced Learning Technologies (ICALT), 2021, pp. 176178.

[4] S. Lee-Cultura, K. Sharma, and M. N. Giannakos, Multimodal Teacher Dashboards: Challenges and Opportunities of Enhancing Teacher Insights through a Case Study, IEEE Trans. Learning Technologies, vol. 17, no. 2, pp. 181201, 2024.

[5] M. Ali, T. Baker and Y. Huang, Offline-first mobile applications: Design and implementation challenges, IEEE Pervasive Computing, vol. 18, no. 3, pp. 14 22, 2019.

[6] A. Kebede and S. Bhattacharya, Student Engagement Awareness Dashboard to Asynchronous E-learning Environment, in ICT Systems and Sustainability, LNNS, vol. 321, Springer, 2022, pp. 759771.

[7] V. A., K. Bharath, and T. N. Anitha, A Novel Video-on-Demand System Architecture to E-Learning in Rural Regions of India, in Proc. Int. Conf. Computational Intelligence and Applications (ICCIA), 2012, pp. 1 5.

[8] Education for All Global Monitoring Report, Paris, France: UNESCO, 2020.

[9] World Bank, World Development Report 2018: Learning to Realize Education Promises, 2018, Washington, DC, USA.

[10] J. Brown, A. Smith, and L. White, "Bridging the digital divide: ICT adoption in underserved communities," IEEE Internet Computing, vol. 24, no. 5, pp. 4450, 2020.

[11] Fernandez and J. Gomez, Community-based ICT models of blended learning in Latin America Educational Technology Research Journal, vol. 28, no. 3, pp. 5667, 2021.

[12] R. Agarwal, A. Joshi, and M. Kumar, "ICT interventions in enhancing rural school education in India: a case study, IEEE Global Humanitarian Technology Conf. (GHTC), 2019, pp. 16.

[13] K. A. Pathak and V. Kumar, Design of an e-learning framework in resource-constrained environments, IEEE Int. Conf. Technology for Education (T4E), 2018, pp. 42 49.

[14] A. Kumar, R. Mishra, and S. Sharma, Improving access to e-learning systems with the help of multilingual content, IEEE Access, vol. 8, pp. 2114521153, 2020.

[15] B. Prakash and P. Srinivasan, A comparative study of online and offline e-learning models in rural education, Int. Conf. Communication Systems and Networks (COMSNETS), 2020, pp. 650655.

[16] P. Chaudhary and K. Raj, Location: Role of cloud computing in transforming e-learning in developing nations, IEEE Int. Conf. Cloud Computing in Emerging Markets (CCEM), 2019, pp. 112118.

[17] H. W. Goh, J. L. Lee, P. C. Chua, (2021), Mobile technologies in rural education A usability study, IEEE Int. Conf. Mobile Learning, p. 199204.

[18] R. S. Yadav and S. Sharma, "Design of an offline-first framework of rural e-learning applications, Proc. IEEE Int. Conf. Advanced Networks and Telecommunications Systems (ANTS), 2021, pp. 153158.

[19] UNESCO Institute of statistics, Meeting commitments: Are countries on track towards education SDG 4 Paris, France: UNESCO, 2019.

[20] OECD, The digital education outlook 2021: Paving the way to future learning, OECD publishing, Paris, 2021.

Base Paper

R. Shaw, K. Shiwaku, Y. Kobayashi and M. Kobayashi,
"Connecting Experience, Education, Perception and Earthquake Preparedness,
Disaster Prevention and Management: An International Journal, vol., 13, no. 1, pages 39-49, 2004.

Justification

The significance of disaster education in enhancing preparedness and awareness of students and communities has been identified in this paper. It highlights that well organized learning programs and training in schools can greatly improve the capacity of the individuals to respond appropriately in times of disasters. This concept has inspired the suggested Disaster Preparedness and Response Education System as it aims to equip teachers with digital platform and interactive learning modules, virtual disaster drills, and guidance tools that will enhance disaster awareness and preparedness in educational establishments.

APPENDIX A – Dataset Details and Accessibility

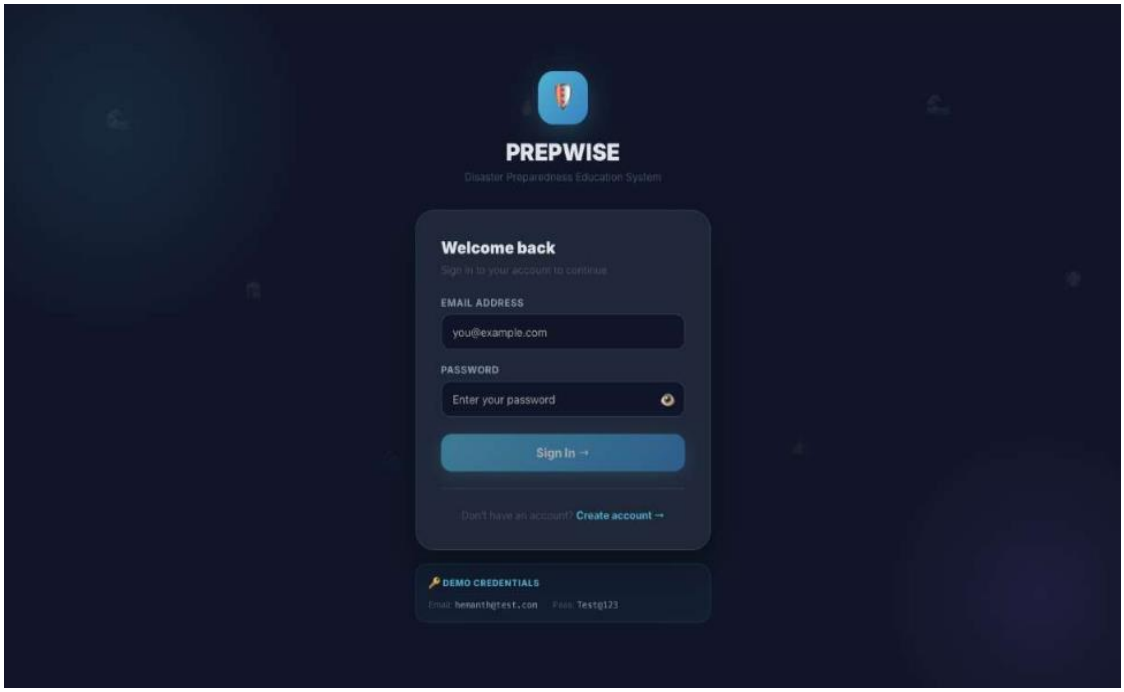
The Global Disaster Information Dataset is available through the EM-DAT (Emergency Events Database), maintained by the Centre for Research on the Epidemiology of Disasters. It is a comprehensive, publicly accessible dataset containing detailed records of global disaster events used for research, policy-making, and disaster preparedness analysis.The dataset includes over 22,000 records covering various types of natural and technological disasters worldwide. It provides structured information such as disaster type, location, impact, and temporal details, making it suitable for disaster management and emergency preparedness studies.

Dataset Source: NASA Prognostics Data Repository

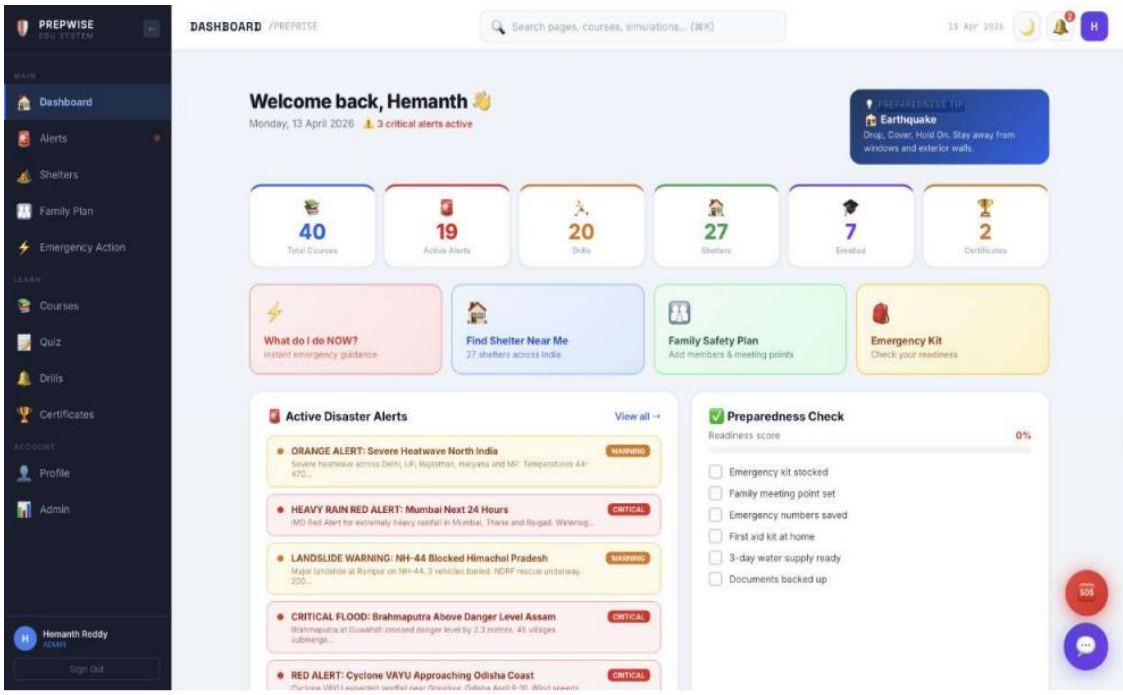
Access URL: https://www.emdat.be/

APPENDIX B - Images Related To The Project

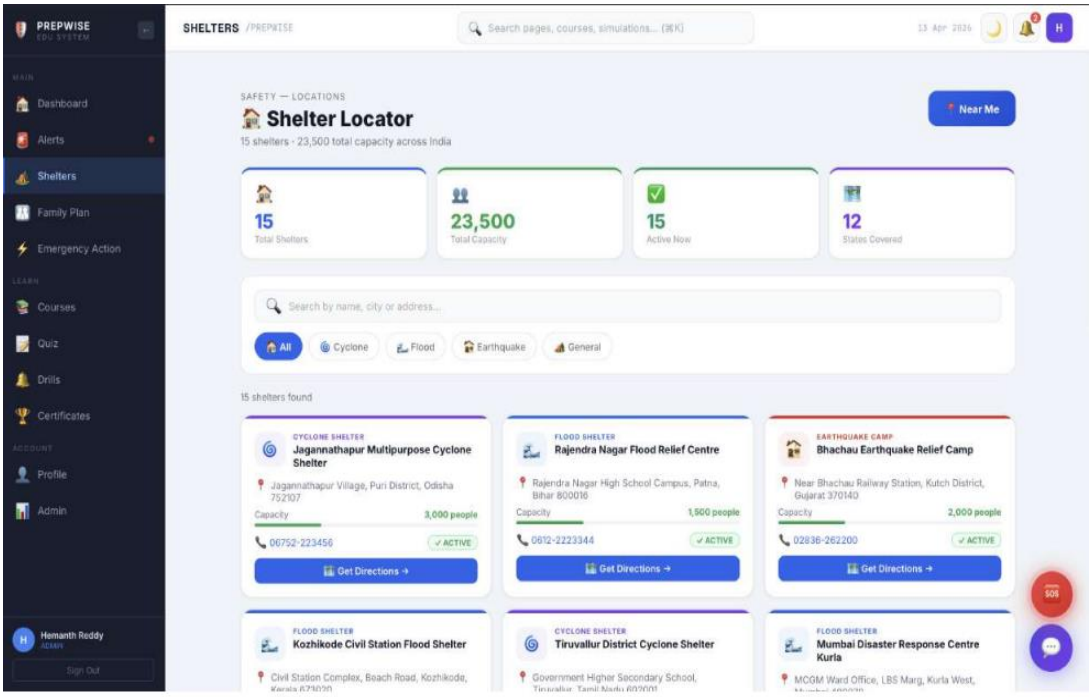
1.Login page



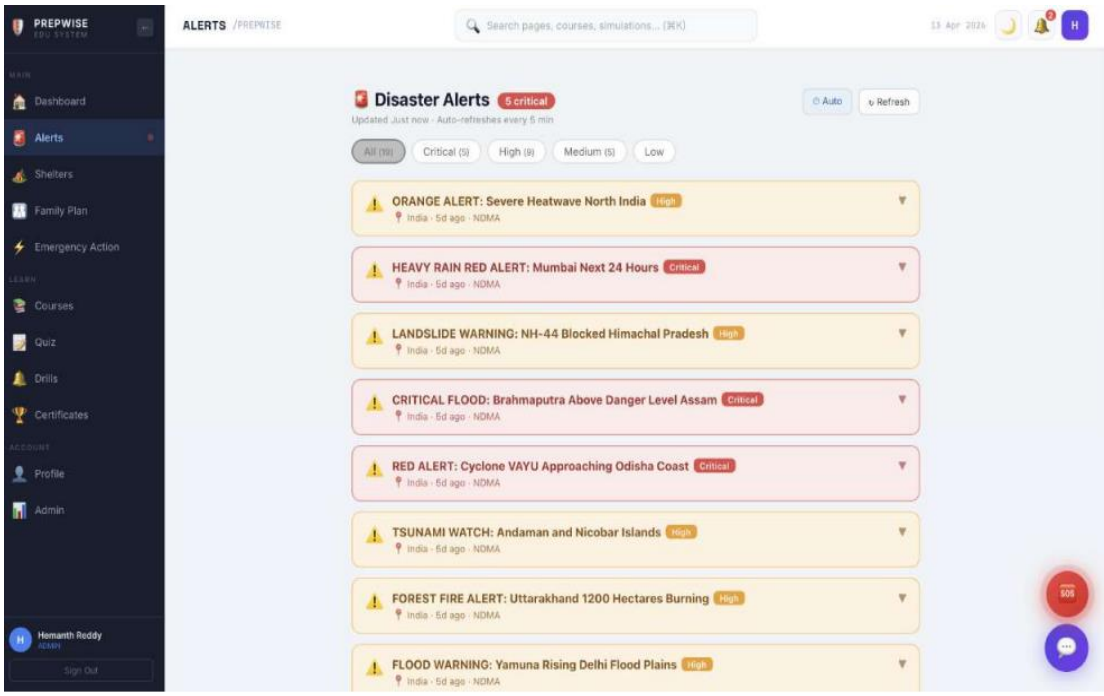
2.Dashboard



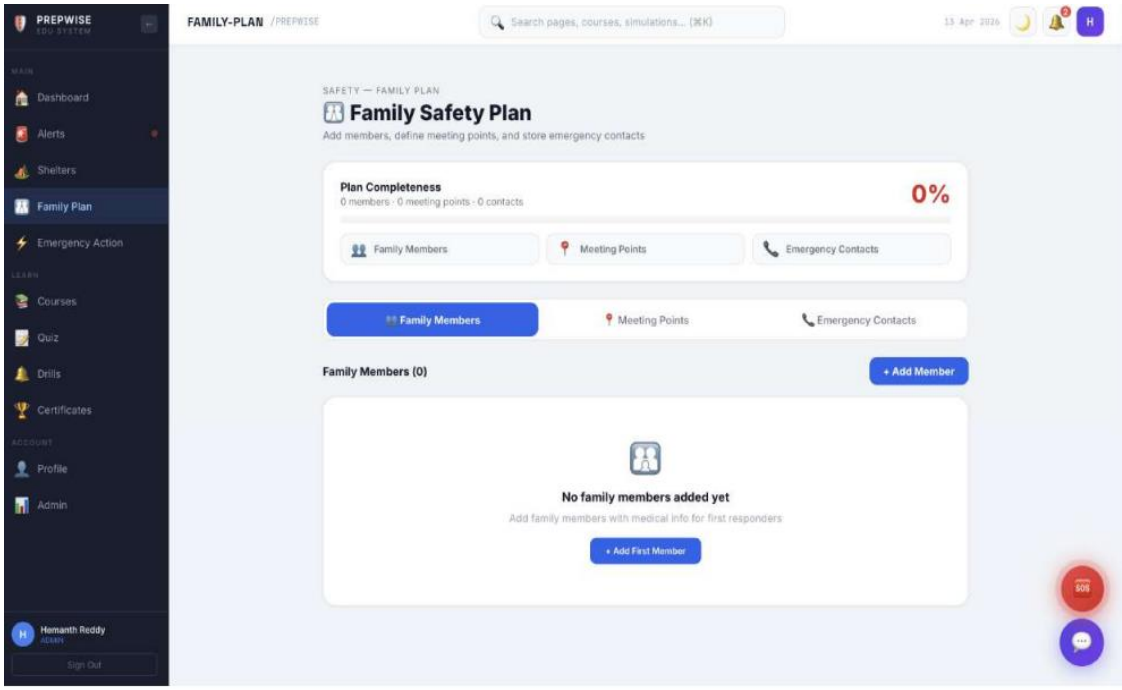
3.Shelter Locator



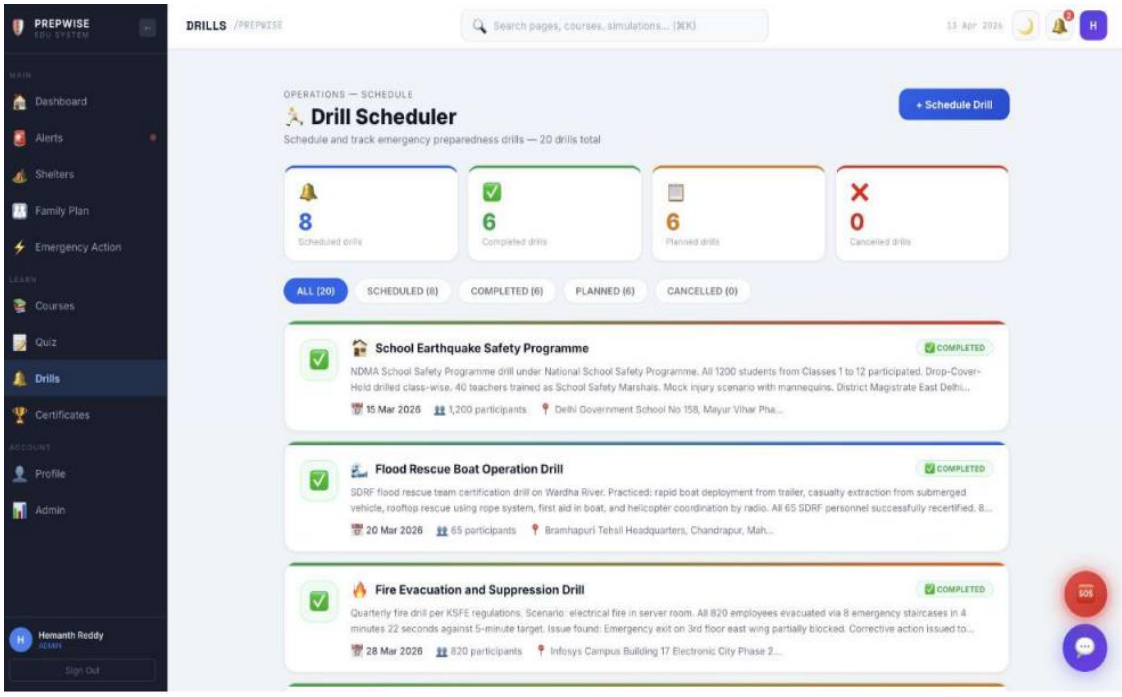
5.Alert Page



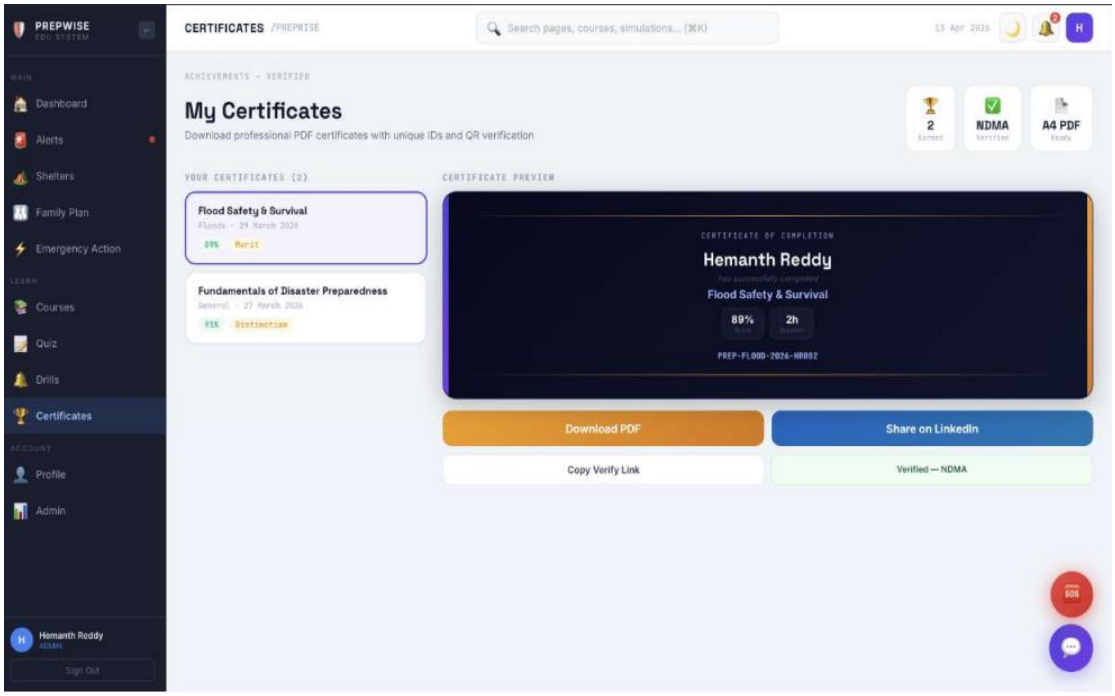
6.Family Safety Plan



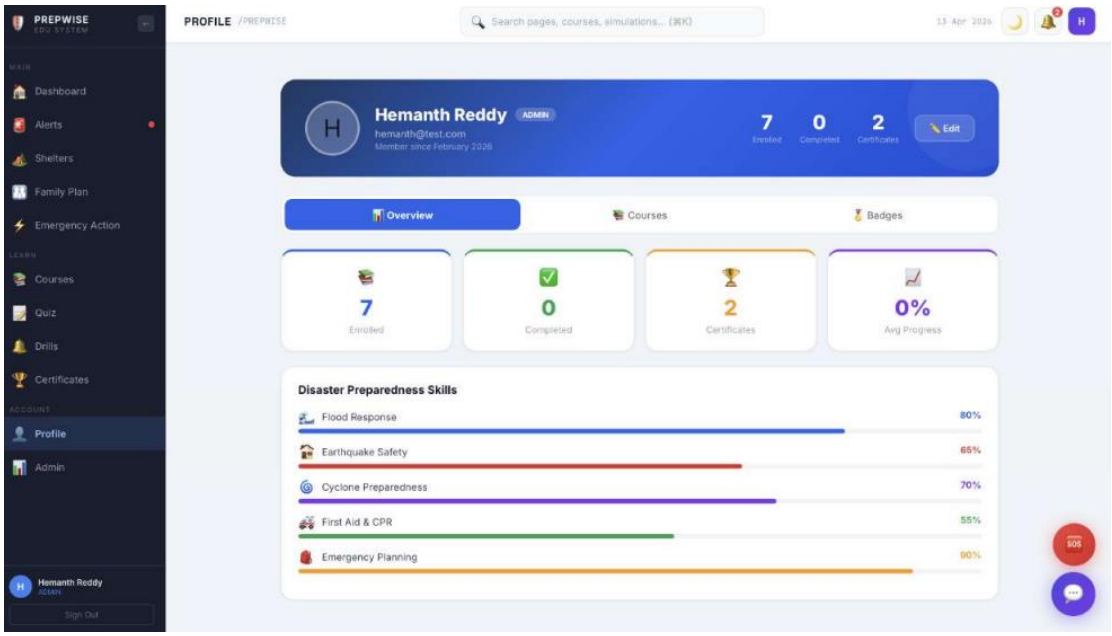
7.Drill Scheduler



8.Certificates



9. Profile



APPENDIX C - Research Paper Submission Acknowledgment

2026 International Conference on AI-Enabled Smart Solutions for Industry, Governance, Healthcare, and Sustainable Development : Submission (135) has been created. Inbox ×

Microsoft CMT <noreply@msr-cmt.org> 12:53 PM (44 minutes ago)
to me

Hello,

The following submission has been created.

Track Name: AISIGHSD2026

Paper ID: 135

Paper Title: Disaster Preparedness and Response Education System for Schools and Colleges

Abstract:
Natural catastrophes like floods, earthquakes, cyclones and landslides, still remain a great threat to the societies all over the world. Poor preparedness training (particularly to the students and the youth overall) is one of the biggest challenges in disaster management today. The proposed study aims to solve this problem by coming up with an education system in Disaster Preparedness and Response (DPRES). This system aims to enhance awareness, knowledge and general preparedness using a captivating online learning system.The site offers interactive educational materials, virtual disaster simulation and quizzes in the form of games to make the learning process more exciting and efficient. These characteristics enable the user to have a better idea on how to react in case of an emergency. Moreover, the system will have a chatbot based on AI, which may provide real-time assistance, respond to the questions connected with disaster safety, and inform users about what they should do in various emergency cases.. The system also has an administrative dashboard which facilitates the institutions to check the participation of students and measure the level of readiness, and to control the educational content effectively. Data on all the systems, such as user activities and simulating results, are stored in a secure cloud based data base so that it is scalable, can be accessed and guaranteed to perform well. The proposed platform will enhance disaster preparedness education and encourage

APPENDIX D - Research Paper

Disaster Preparedness and Response Education System for Schools and Colleges

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Abstract—*Natural catastrophes like floods, earthquakes, cyclones and landslides, still remain a great threat to the societies all over the world. Poor preparedness training (particularly to the students and the youth overall) is one of the biggest challenges in disaster management today. The proposed study aims to solve this problem by coming up with an education system in Disaster Preparedness and Response (DPRES). This system aims to enhance awareness, knowledge and general preparedness using a captivating online learning system.The site offers interactive educational materials, virtual disaster simulation and quizzes in the form of games to make the learning process more exciting and efficient. These characteristics enable the user to have a better idea on how to react in case of an emergency. Moreover, the system will have a chatbot based on AI, which may provide real-time assistance, respond to the questions connected with disaster safety, and inform users about what they should do in various emergency cases.. The system also has an administrative dashboard which facilitates the institutions to check the participation of students and measure the level of readiness, and to control the educational content effectively. Data on all the systems, such as user activities and simulating results, are stored in a secure cloud based data base so that it is scalable, can be accessed and guaranteed to perform well. The proposed platform will enhance disaster preparedness education and encourage users to act in a proactive manner in terms of safety behaviors by combining the use of interactive learning tools and intelligent assistance. The findings indicate that technological-supported education can play a great role in increasing disaster awareness and preparedness in academic*

Keywords— *Flood Awareness, Disaster preparedness, e-learning, AI chatbot, virtual drills, cloud computing, gamification, educational technology, emergency response, and NDMA.*

I. Introduction

India is one of the most disaster prone countries because of its geographical diversity, seismic activity, monsoon dependence and the high population. The schools and colleges are especially susceptible settings since they are the places where masses of young people can be found, who might lack proper knowledge or experience to act in a proper manner in case of an emergency. Clustering of the students in the limited space of the campus heightens the risk that may occur in the event of disasters like earthquakes, fire, flood

Despite disaster management being identified as one of the national priorities, there is still a low integration of organized disaster education in formal education. Most of the institutions carry out awareness only on a few occasions and have no hands-on training, simulations and systematic assessment of preparedness levels.

Consequently, students and staff might get disoriented, panic and make wrong decisions during actual emergencies. Moreover, institutions have no standardized digital means to regularly monitor reliance and assess training program orientation, which is why the introduction of more digital tools is a promising chance to update disaster education based on interactive learning tools. Scalable and accessible disaster training can be facilitated by web-based systems, cloud infrastructure and real-time communication technologies.

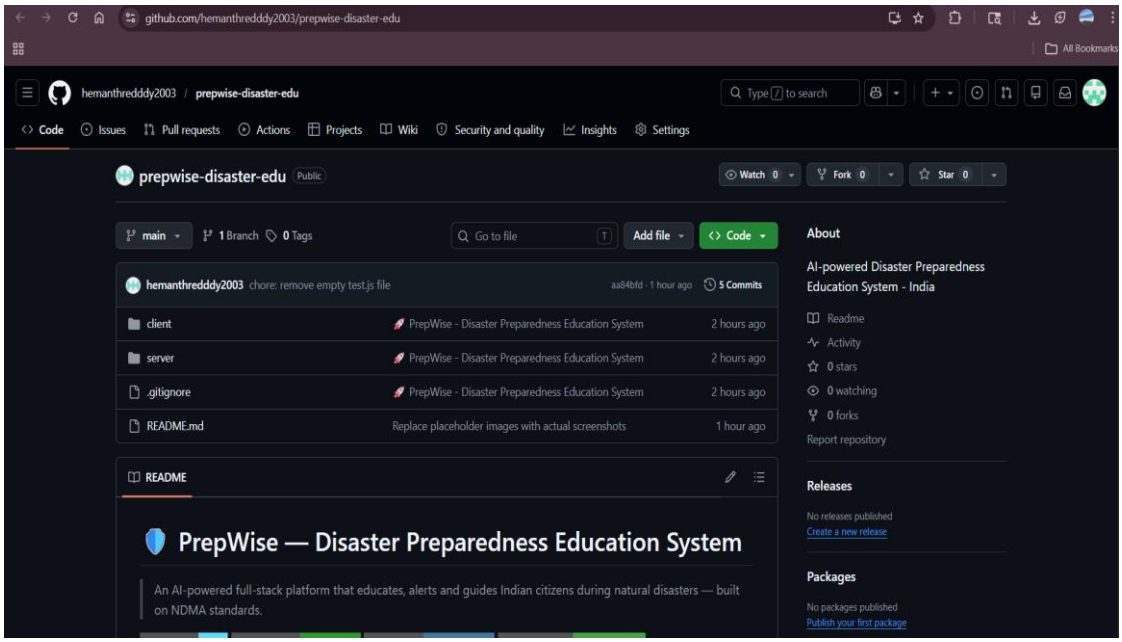
This paper suggests a Digital Disaster Preparedness and Response Education System (DPRES) which makes use of web technologies, cloud computing, and gamification to offer continuous, interactive and realistic disaster education. The platform is also not only educative to the user, but it helps the institutions to track readiness, measure performance, produce preparedness reports, and ensure safety guidelines set are being adhered to. The proposed system will help to enhance institutional resilience and create a spirit of proactive disaster preparedness by integrating education, simulation, and analytics into one system.

II. Literature Survey

TABLE I. Summarizes The Major Research Works On Disaster Preparedness Education That is Technology Driven And The Findings

Author & Year	Method /Approach	Key findings
Johnson et al.(2022)	Online disaster education courses	The level of preparedness increased by approximately 65 percent as opposed to the old lecture techniques

APPENDIX E - Source Code Repository

GitHub Link

https://github.com/hemanthredddy2003/prepwise-disaster-edu